Bayesian Penalized Transformation Models

Appendix

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A. Transformation function details

The full transformation function is given by

$$h(r) = \begin{cases} r + \tilde{A} & \text{if} \quad r \in (-\infty, a-\lambda) & \text{(linear)}, \\ \text{left}(r) + A & \text{if} \quad r \in [a-\lambda, a) & \text{(transition)}, \\ \text{spline}(r) & \text{if} \quad r \in [a, b] & \text{(core)}, \\ \text{right}(r) + B & \text{if} \quad r \in (b, b+\lambda] & \text{(transition)}, \\ r + \tilde{B} & \text{if} \quad r \in (b+\lambda, \infty) & \text{(linear)}, \end{cases} \tag{1}$$

where $\lambda \in \mathbb{R}_{>0}$, $a < b$, and $a, b, \tilde{A}, A, \tilde{B}, B \in \mathbb{R}$. The core segment is a modified increasing B-spline with real-valued parameters $\boldsymbol{\delta}^{\mathsf{T}} = [\delta_1, \ldots, \delta_{J-1}]$ given by

$$\text{spline}(r) = \alpha + \underbrace{\frac{1}{s(\boldsymbol{\delta})} \sum_{j=2}^{J} B_j(r) \left(\sum_{\ell=2}^{j} \exp(\delta_{\ell-1}) \right)}_{:=g(r)}, \tag{2}$$

where $B_j(r)$ is a third-order (cubic) B-spline basis evaluation. The bases are set up using $J+4$ equidistant knots $k_{-2} < k_{-1} < k_0 < \cdots < k_J < k_{J+1}$, where $k_1, \ldots, k_m$ with $m = J-2$ are called the interior knots and the distance between adjacent knots is $d = k_{j+1} - k_j$. The boundary knots are $k_1 = a$ and $k_m = b$. The intercept parameter α is given by $\alpha = a - g(a)$. The function $s(\boldsymbol{\delta})$ is given by

$$s(\boldsymbol{\delta}) = \frac{1}{b-a} \sum_{j=1}^{J-3} \left(\frac{\exp(\delta_j) + \exp(\delta_{j+2})}{6} + \frac{2\exp(\delta_{j+1})}{3} \right). \tag{3}$$

The left and right transition segments provide smooth transitions to the linear extrapolation segments. They are given by

$$\text{left}(r) = r \left[\frac{a - r/2}{\lambda} \right] + r \left[1 - \frac{a - r/2}{\lambda} \right] \frac{\partial}{\partial a} \text{spline}(a), \text{ and} \tag{4}$$

$$\text{right}(r) = r \left[\frac{r/2 - b}{\lambda} \right] + r \left[1 - \frac{r/2 - b}{\lambda} \right] \frac{\partial}{\partial b} \text{spline}(b). \tag{5}$$

The constants $A = \text{spline}(a) - \text{left}(a)$ and $B = \text{spline}(b) - \text{right}(b)$ in the transition segments induce vertical shifts, ensuring that the function values at the segment boundaries align. The constant shifts in the linear function segments are $\tilde{A} = \text{left}(a-\lambda) - (a-\lambda) + A$ and $\tilde{B} = \text{right}(b+\lambda) - (b+\lambda) + B$.

B. Proofs

We first use Definition B.1 to define a monotonically increasing B-spline.

Definition B.1 (B-spline)**.** Let $f : \mathbb{R} \to \mathbb{R}$ be given by

$$f(r) = \sum_{j=1}^{J} B_j^{(3)}(r)\omega_j, \tag{6}$$

with spline coefficients $\omega_1, \dots, \omega_J \in \mathbb{R}$ and B-spline bases $B_j^{(3)}$ of order 3 using an increasing knot base $k_{-2} < k_{-1} < k_0 < \cdots < k_J < k_{J+1}$. Then we call (6) a B-spline.

We use the symbol f to denote the function here instead of h as used in the main text to distinguish between a pure spline and the full transformation function h. The knots $k_1, \dots, k_m$ with $m = J - 2$ are called the interior knots. We are working exclusively with equidistant knots, and denote the distance between adjacent knots as $d = k_{j+1} - k_j$. The boundary knots are $k_1 = a$ and $k_m = b$.

We use the recursive definition of a B-spline basis as given by Fahrmeir et al. (2013):

$$B_j^{(p)}(r) = \frac{r - k_{j-p}}{k_j - k_{j-p}} B_{j-1}^{(p-1)}(r) + \frac{k_{j+1} - r}{k_{j+1} - k_{j+1-p}} B_j^{(p-1)}(r). \tag{7}$$

The order of the basis is given by p, with $p = 3$ for cubic splines. The basis of order $p = 0$ is given by

$$B_j^{(0)}(r) = I(k_j \leq r < k_{j+1}), \qquad j = 1, \dots, J-1,$$

where $I(\cdot)$ is the indicator function for the enclosed condition. For equidistant knots $k_{j+1} - k_j = d$, the basis (7) simplifies to

$$B_j^{(p)}(r) = \frac{r - k_{j-p}}{pd} B_{j-1}^{(p-1)}(r) + \frac{k_{j+1} - r}{pd} B_j^{(p-1)}(r). \tag{8}$$

B.1. Proposition B.1 (Derivative)

Proposition B.1. *The first derivative of h with respect to r is given by*

$$\frac{\partial}{\partial r} h(r) = \frac{1}{s(\boldsymbol{\delta}) \cdot d} \sum_{j=2}^{J} B_j^{(2)}(r) \exp(\delta_{j-1}),$$

where $d = k_{j+1} - k_j$ is the (constant) distance between adjacent knots and $B_j^{(2)}(r)$ is a quadratic B-spline basis function evaluation.

Proof. The proposition is concerned with the derivative of the transformation function's core segment, which is given by

$$h(r) = \alpha + \frac{1}{s(\boldsymbol{\delta})} \sum_{j=1}^{J} B_j(r) \sum_{\ell=2}^{j} \exp(\delta_{\ell-1}) \quad \text{if} \quad r \in [a, b],$$

where B_j are cubic B-spline bases. The function can be written as a B-spline $h(r) = s(\boldsymbol{\delta})^{-1}\tilde{h}(r)$, where $\tilde{h}(r) = \sum_{j=1}^{J} B_j(r)\omega_j$ is a cubic B-spline with spline coefficients defined as $\omega_j = \alpha +$

$\sum_{\ell=2}^{j} \exp(\delta_{\ell-1})$. It is a well-known property of B-splines that the derivative of a B-spline of order p is another B-spline of order $p-1$, with the spline coefficients given by differences in the original coefficients divided by the distance between the corresponding knots (see, for example, Eilers & Marx, 2021, p. 20). Since we use equidistant knots, this distance is the constant d in our case. Since we use a B-spline of order $p = 3$, the derivative of $\tilde{h}$ is a B-spline of order $p = 2$, given by

$$\frac{\partial}{\partial r}\tilde{h}(r) = \sum_{j=2}^{J} B_j^{(2)}(r)\frac{\omega_j - \omega_{j-1}}{d} = \frac{1}{d}\sum_{j=2}^{J} B_j^{(2)}(r)\exp(\delta_{-1}).$$

Combining this partial result with the scaling by $s(\boldsymbol{\delta})^{-1}$, we arrive at the result given in Proposition B.1.

□

B.2. Proof of Proposition 2.1 (Monotonicity)

To prove this proposition, we use $h : \mathbb{R} \to \mathbb{R}$ and the core segment spline(r) as defined fully in Appendix A. Note that the derivative of this transformation function is

$$\frac{\partial}{\partial r}h(r) = \begin{cases} 1 & \text{if} \quad r \in (-\infty, a-\lambda) \qquad \text{(linear)}, \\ \left(1 - \frac{a-r}{\lambda}\right) \cdot \frac{\partial}{\partial a}\text{spline}(a) + \frac{a-r}{\lambda} & \text{if} \quad r \in [a-\lambda, a) \qquad \text{(transition)}, \\ \frac{\partial}{\partial r}\text{spline}(r) & \text{if} \quad r \in [a, b] \qquad \text{(core)}, \\ \left(1 - \frac{r-b}{\lambda}\right) \cdot \frac{\partial}{\partial b}\text{spline}(b) + \frac{r-b}{\lambda} & \text{if} \quad r \in (b, b+\lambda] \qquad \text{(transition)}, \\ 1 & \text{if} \quad r \in (b+\lambda, \infty) \qquad \text{(linear)}. \end{cases} \tag{9}$$

Proof. We prove the theorem by showing that $\frac{\partial}{\partial r}h(r) > 0$ for all $r \in \mathbb{R}$, which is equivalent to strict monotonicity. We proceed by cases.

1. *Case 1: Linear segments.* Note that in the linear function segments, when $r \in (-\infty, a - \lambda) \cup (b + \lambda, \infty)$, we have $\frac{\partial}{\partial r}h(r) = 1$, which is strictly greater than zero. Thus, in the linear segments, h is strictly monotonically increasing.

2. *Case 2: Core segment.* By Proposition B.1, the derivative in the core segment when $r \in [a, b]$ is given by

$$\frac{\partial}{\partial r}\text{spline}(r) = \frac{1}{s(\boldsymbol{\delta}) \cdot d}\sum_{j=2}^{J} B_j^{(2)}(r)\exp(\delta_{j-1}).$$

   Note that, by definition, any basis function evaluation $B_j^{(2)}(r)$ is nonnegative, and for any given $r \in [a, b]$, three basis function evaluations are positive. Note also that the distance between adjacent knots $d = k_{j+1} - k_j > 0$ by construction. Finally, observe that $\exp(\delta) > 0$ for all $\delta \in \mathbb{R}$, and consequently $s(\boldsymbol{\delta}) > 0$. As a result, $\frac{\partial}{\partial r}\text{spline}(r) > 0$ for all $r \in [a, b]$.

3. *Case 3: Transition segments.* We start with the left transition segment. The derivative of this segment is given by

$$\frac{\partial}{\partial r}\,\text{left}(r) = \left(1 - \frac{a-r}{\lambda}\right) \cdot \frac{\partial}{\partial a}\text{spline}(a) + \frac{a-r}{\lambda}$$

Observe that $a - \lambda \leq r < a$ and $\lambda > 0$, such that $\frac{a-r}{\lambda} \in (0, 1]$. Thus,

$$\min\left\{\frac{\partial}{\partial r}\,\text{left}(r) : r \in [a-\lambda, a)\right\} = \min\left\{\frac{\partial}{\partial a}\text{spline}(a), 1\right\}.$$

Since we know that $\frac{\partial}{\partial r}\text{spline}(r) > 0$ for all $r \in [a, b]$ from Case 2, this means that $\frac{\partial}{\partial r}\,\text{left}(r) > 0$ for all $r \in [a-\lambda, a]$. The proof for the right transition, when $r \in (b, b+\lambda]$, works analogously.

We have shown that $\frac{\partial}{\partial r}h(r) > 0$ if $r \in (-\infty, a-\lambda) \cup (b+\lambda, \infty)$ (Case 1), if $r \in [a-\lambda, a)$ or $r \in (b, b+\lambda]$ (Case 3), and if $r \in [a, b]$ (Case 2). Observe that

$$\mathbb{R} = (-\infty, a-\lambda) \cup [a-\lambda, a) \cup [a, b] \cup (b, b+\lambda] \cup (b+\lambda, \infty).$$

Thus, we conclude that $\frac{\partial}{\partial r}h(r) > 0$ for all $r \in \mathbb{R}$, which is equivalent to saying that h is strictly monotonically increasing in r for all $r \in \mathbb{R}$.

□

B.3. Proof of Theorem 2.1 (Core probability)

This proof has a longer setup. First, we derive the average slope of a monotonically increasing B-spline in Proposition B.2 with the help of the Lemmas B.1 and B.2. Second, we use Proposition B.2 to prove that the average slope of the transformation function h over the subdomain $[a, b]$ is one in Lemma B.3. Finally, we prove the theorem by showing first that $h(a) = a$, then using Lemma B.3 to show that $h(b) = b$.

Lemma B.1 (Derivative in a segment)**.** *Let $f(r) = \sum_{j=1}^{J} B_j(r)\omega_j$ be a monotonically increasing cubic B-spline (see Definition B.1) with spline coefficients $\omega_j = \alpha + \sum_{\ell=2}^{j} \exp(\delta_{\ell-1})$, where $\alpha, \delta_1, \delta_2, \ldots, \delta_{J-1} \in \mathbb{R}$. If $r \in [k_j, k_{j+1}]$, where $1 \leq j \leq J-3$, then the first derivative of f evaluated at r is*

$$\frac{\partial}{\partial r}f(r) = \sum_{j'=j}^{j+2} B_{j'}^{(2)}(r)\frac{\exp(\delta_{j'})}{d}. \tag{10}$$

Proof. A well-known property of B-splines of order p is the fact that at any point $\varepsilon \in [k_1, k_{J-2}]$, we have $p+1$ positive basis functions, while the remaining basis functions evaluate to zero (see Fahrmeir et al., 2013, p. 429). Since the derivative of f uses bases of order $p = 2$, that means at any point $r \in [k_1, k_{J-2}]$, we have 3 positive basis functions. If we know that $r \in [k_j, k_{j+1}]$, it follows from the definition of B-spline bases (8) that the positive basis functions are B_j, B_{j+1}, and B_{j+2}, which directly lets the general form of the derivative simplify to (10).

$\square$

Lemma B.2 (Average derivative in a segment)**.** *Let $f(r) = \sum_{j=1}^{J} B_j(r)\omega_j$ be a monotonically increasing cubic B-spline (see Definition B.1) with spline coefficients $\omega_j = \alpha + \sum_{\ell=2}^{j} \exp(\delta_{\ell-1})$, where $\alpha, \delta_1, \delta_2, \ldots, \delta_{J-1} \in \mathbb{R}$. If $r \in [k_j, k_{j+1}]$, the average derivative of f in the corresponding function segment is given by*

$$\frac{1}{d} \int\limits_{k_j}^{k_{j+1}} \left(\frac{\partial}{\partial r} f(r) \right) \mathrm{d}r = \frac{1}{d} \left(\frac{1}{6} \exp(\delta_j) + \frac{2}{3} \exp(\delta_{j+1}) + \frac{1}{6} \exp(\delta_{j+2}) \right). \tag{11}$$

Proof. First, note that since $r \in [k_j, k_{j+1}]$, we can apply Lemma B.1 to write

$$\frac{1}{d} \int\limits_{k_j}^{k_{j+1}} \left(\frac{\partial}{\partial r} f(r) \right) \mathrm{d}r = \frac{1}{d} \left(\frac{\exp(\delta_j)}{d} \int\limits_{k_j}^{k_{j+1}} B_j^{(2)}(r)\mathrm{d}r + \frac{\exp(\delta_{j+1})}{d} \int\limits_{k_j}^{k_{j+1}} B_{j+1}^{(2)}(r)\mathrm{d}r + \frac{\exp(\delta_{j+2})}{d} \int\limits_{k_j}^{k_{j+1}} B_{j+2}^{(2)}(r)\mathrm{d}r \right).$$

A sufficient condition for this expression to simplify to (11) is given by the following three statements:

$$\int\limits_{k_j}^{k_{j+1}} B_j^{(2)}(r)\mathrm{d}r \stackrel{!}{=} \frac{d}{6}, \tag{s1}$$

$$\int\limits_{k_j}^{k_{j+1}} B_{j+1}^{(2)}(r)\mathrm{d}r \stackrel{!}{=} \frac{2d}{3}, \quad \text{and} \tag{s2}$$

$$\int\limits_{k_j}^{k_{j+1}} B_{j+2}^{(2)}(r)\mathrm{d}r \stackrel{!}{=} \frac{d}{6}. \tag{s3}$$

In the following we show first that s1 and s3 hold, before showing that s2 holds.

1. To show that s1 holds, we substitute the definition of a B-spline basis (8) and simplify, such that we obtain

$$\int\limits_{k_j}^{k_{j+1}} B_j^{(2)}(r)\mathrm{d}r = \frac{1}{2d^2} \int\limits_{k_j}^{k_{j+1}} (k_{j+1} - r)^2 \mathrm{d}r.$$

   The integral is readily solved using u-substitution with $u = k_{j+1} - r$. Inserting the integral limits, we get

$$\frac{1}{2d^2} \int\limits_{k_j}^{k_{j+1}} (k_{j+1} - r)^2 \mathrm{d}r = \frac{1}{2d^2} \frac{(k_{j+1} - k_j)^3}{3} = \frac{d^3}{6d^2} = \frac{d}{6},$$

   confirming that s1 is true. The solution for s3 works in complete analogy.

2. Substituting the definition of a B-spline basis into s2, we obtain

$$\int_{k_j}^{k_{j+1}} B_{j+1}^{(2)}(r)\mathrm{d}r = \frac{1}{2d^2}\left(-\int_{k_j}^{k_{j+1}} (r-k_{j-1})(r-k_{j+1})\mathrm{d}r - \int_{k_j}^{k_{j+1}} (r-k_j)(r-k_{j+2})\mathrm{d}r\right). \quad (12)$$

The two integrals in (12) can be solved analogously. Considering the left-hand term, we have

$$-\int_{k_j}^{k_{j+1}} (r-k_{j-1})(r-k_{j+1})\mathrm{d}r = -\left[\left(\frac{2k_{j+1}^3 - 6k_{j+1}^2 d}{6}\right) - \left(\frac{-4k_j^3 + 6k_j k_{j-1} k_{j+1}}{6}\right)\right].$$

Next, we substitute $k_j = k_{j+1} - d$ and $k_{j-1} = k_{j+1} - 2d$ and simplify further, which results in

$$-\int_{k_j}^{k_{j+1}} (r-k_{j-1})(r-k_{j+1})\mathrm{d}r = \frac{2d^3}{3}.$$

Applying the analogous procedure to the right-hand term in (12), and inserting both parts back in (12), we get

$$\int_{k_j}^{k_{j+1}} B_{j+1}^{(2)}(r)\mathrm{d}r = \frac{1}{2d^2}\left(\frac{2d^3}{3} + \frac{2d^3}{3}\right) = \frac{2d}{3},$$

concluding the proof of s2.

Since we have shown that the sufficient conditions s1, s2, and s3 are met, this concludes the proof.

□

Proposition B.2 (Average slope of a monotonically increasing B-spline). *Let $f(r) = \sum_{j=1}^{J} B_j(r)\omega_j$ be a monotonically increasing cubic B-spline (see Definition B.1) with spline coefficients $\omega_j = \alpha + \sum_{\ell=2}^{j} \exp(\delta_{\ell-1})$, where $\alpha, \delta_1, \delta_2, \ldots, \delta_{J-1} \in \mathbb{R}$. Let $a = k_1$ and $b = k_m$ denote the boundary knots. Then the average slope of f over the interval $[a, b]$ is given by*

$$s(\boldsymbol{\delta}) = \frac{1}{b-a}\int_a^b \frac{\partial}{\partial r} f(r)\mathrm{d}r = \frac{1}{b-a}\sum_{j=1}^{J-3}\left(\frac{\exp(\delta_j) + \exp(\delta_{j+2})}{6} + \frac{2\exp(\delta_{j+1})}{3}\right), \quad (13)$$

Proof. The integral can be additively decomposed using the interior knots as the bounds of individual integrals, such that we can write

$$\frac{1}{b-a}\int_a^b \frac{\partial}{\partial r} f(r)\mathrm{d}r = \frac{1}{b-a}\sum_{j=1}^{J-3}\left(\int_{k_j}^{k_{j+1}} \frac{\partial}{\partial r} f(r)\mathrm{d}r\right).$$

By Lemma B.2, the individual integrals in the right-hand-side sum can be written in terms of the parameters δ_j, δ_{j+1}, and δ_{j+2}, such that we can write

$$\frac{1}{b-a}\int\limits_a^b \frac{\partial}{\partial r} f(r)\mathrm{d}r = \frac{1}{b-a}\sum_{j=1}^{J-3}\left(\frac{1}{6}\exp(\delta_j) + \frac{2}{3}\exp(\delta_{j+1}) + \frac{1}{6}\exp(\delta_{j+2})\right),$$

which directly simplifies to (13).

□

Lemma B.3 (Average slope over core segment)**.** *The average slope of the transformation function* $h : \mathbb{R} \to \mathbb{R}$ *as defined in Appendix A over the core segment* $[a, b]$ *is one.*

Proof. Observe that the derivative of h with respect to r for $r \in [a, b]$ is identified in Proposition B.1 as

$$\frac{\partial}{\partial r} h(r) = \frac{1}{s(\boldsymbol{\delta}) \cdot d}\sum_{j=2}^{J} B_j^{(2)}(r)\exp(\delta_{j-1}).$$

Note further that h can be written as a B-spline $h(r) = s(\boldsymbol{\delta})^{-1}\tilde{h}(r)$, where $\tilde{h}(r) = \sum_{j=1}^{J} B_j(r)\omega_j$ is a cubic B-spline with spline coefficients defined as $\omega_j = \alpha + \sum_{\ell=2}^{j}\exp(\delta_{\ell-1})$. As a result, the derivative of h can be written as

$$\frac{\partial}{\partial r} h(r) = \frac{1}{s(\boldsymbol{\delta})}\frac{\partial}{\partial r}\tilde{h}(r).$$

Since by Proposition B.2, $s(\boldsymbol{\delta})$ is the average derivative of $\tilde{h}$ over $[a, b]$, we can see that dividing by $s(\boldsymbol{\delta})$ normalizes the average slope of h over $[a, b]$ to one:

$$\frac{1}{b-a}\int_a^b \frac{\partial}{\partial r} h(r)\mathrm{d}r = \frac{1}{s(\boldsymbol{\delta})}\frac{1}{b-a}\int_a^b \frac{\partial}{\partial r}\tilde{h}(r)\mathrm{d}r = \frac{s(\boldsymbol{\delta})}{s(\boldsymbol{\delta})} = 1.$$

□

Main proof of Theorem 2.1

We use the transformation function $h : \mathbb{R} \to \mathbb{R}$ as defined in Appendix A.

Proof. First, note that $h(a) = a$ by construction, since $\alpha = a - h(a|\alpha = 0)$, where $h(a|\alpha = 0)$ denotes a preliminary evaluation of h where α is temporarily set to 0. Second, note that by Lemma B.3, the average slope of h over $[a, b]$ is one. Since $h(a) = a$, this property directly implies $h(b) = b$. Next, recall that the cumulative distribution function of R is defined as $\mathbb{P}(R \leq r) = F_Z(h(r))$. It follows that $\mathbb{P}(R \leq a) = F_Z(h(a)) = F_Z(a)$ and $\mathbb{P}(R \leq b) = F_Z(h(b)) = F_Z(b)$, concluding the proof. □

B.4. Proof of Theorem 2.2 (Reduction to identity)

We use the transformation function $h : \mathbb{R} \to \mathbb{R}$ as defined in Appendix A.

Proof. First, consider the core segment, where $r \in [a, b]$. If $\delta_j = c$ for all $j = 1, \dots, J-1$, the derivative of h in the core segment can be written as

$$\frac{\partial}{\partial r} h(r) = \frac{\exp(c)}{s(\boldsymbol{\delta}) \cdot d} \sum_{j=2}^{J} B_j^{(2)}(r) = \frac{\exp(c)}{s(\boldsymbol{\delta}) \cdot d},$$

where the second identity relies on the unity decomposition property of B-spline bases (see Fahrmeir et al., 2013, p. 429). Note that the property applies here even though the index starts at $j = 2$, since we write the second-order B-spline in the derivative with indices based on the knots of the parent function, which causes $B_1^{(2)}(r) = 0$. Inserting $\delta_j = c$ for all $j = 1, \dots, J-1$, we further get

$$s(\boldsymbol{\delta}) = \frac{1}{b-a} \sum_{j=1}^{J-3} \left(\frac{\exp(\delta_j) + \exp(\delta_{j+2})}{6} + \frac{2\exp(\delta_{j+1})}{3} \right) = \frac{(J-3)\exp(c)}{b-a}.$$

Since there are $J - 3$ function segments in $[a, b]$, defined by equidistant knots, we can write $b - a = (J-3)d$, where d is the distance between adjacent knots, i.e. the width of each function segment. We can now write

$$\frac{\partial}{\partial r} h(r) = \frac{\exp(c)}{(\exp(c)/d) \cdot d} = 1,$$

which means that if all elements of $\boldsymbol{\delta}$ are equal, h has a unit slope on $[a, b]$. Since we observed in the proof of Theorem 2.1 that $h(a) = a$ and $h(b) = b$, we can conclude that, in this case, h is the identity function on $[a, b]$.

Second, consider the left transition segment as defined in (4). If $\delta_j = c$ for all $j = 1, \dots, J-1$, we have seen that $\frac{\partial}{\partial a}\text{spline}(a) = 1$. Then the left transition segment becomes

$$\text{left}(r) = r \left[\frac{a - r/2}{\lambda} \right] + r \left[1 - \frac{a - r/2}{\lambda} \right] = r.$$

The shift $A = \text{spline}(a) - \text{left}(a)$ thus simplifies to $A = 0$. The case of the right transition segment works analogously. Finally, the shift terms $\tilde{A}$ and $\tilde{B}$ in the linear extrapolation segments likewise simplify to zero, completing the identity core segment and identity transition segments with identity extrapolation segments.

□

B.5. Proposition B.3 (Tail probabilities)

Proposition B.3 (Tail probabilities)**.** *Let R be an absolutely continuous random variable with cumulative distribution function $F_R(r) = F_Z(h(r))$, where $h : \mathbb{R} \to \mathbb{R}$ is given by (1), with* spline(r) *as in (2), and where F_Z is a fully specified continuous cumulative distribution function. Then $F_R(r) = F_Z(r + \tilde{A})$ for $r < a - \lambda$ and $F_R(r) = F_Z(r + \tilde{B})$ for $b + \lambda < r$.*

Proof. We consider first the case $r < a - \lambda$. In this case, $h(r) = r + \tilde{A}$, such that $F_R(r) = F_Z(h(r)) = F_Z(r + \tilde{A})$ for $r < a - \lambda$. The second case works analogously. □

By this proposition, the tails of the distribution of R, where we take the tails to mean $R < (a - \lambda)$ or $(b + \lambda) < R$, are described by shifted versions of the reference CDF. The transition segments provide smooth transitions between the flexible core, in which the shape of R's distribution is governed by $\text{spline}(r)$ and the shifted-reference tails. As $\lambda \to 0$, the transition segments vanish, and the tail probabilities become $F_R(r) \to F_Z(r)$ for both $a < r$ and $r > b$. As can be seen in Figure 1 (bottom left panel) of the main text, this comes at the price of a potentially discontinuous jump in $\frac{\partial}{\partial r} h(r)$ at $r = a$ and $r = b$, which causes problems for gradient-based estimation techniques. On the other hand, as $\lambda \to \infty$, the transformation function h has simple linear extrapolation for $r < a$ and $b < r$, and the tail probabilities become $F_R(r) \to F_Z(r/\text{spline}'(a))$ for $r < a$ and $F_R(r) \to F_Z(r/\text{spline}'(b))$ for $b < r$, where we write $\text{spline}'(\cdot)$ to denote the first derivative of $\text{spline}(\cdot)$ with respect to its input. Linear extrapolation could thus lead to extreme tail behavior, if $\text{spline}'(a)$ or $\text{spline}'(b)$ happen to be estimated to be close to zero or considerably larger than one. The former case would induce heavy tails, while the latter case would induce light tails. We expect little information to be available for the estimation of $\text{spline}'(a)$ and $\text{spline}'(b)$ in many cases; and it is unclear how well information about $\text{spline}'(a)$ and $\text{spline}'(b)$ can be directly used to determine tail probabilities. So with a moderate λ, Proposition B.3 can be thought of as a safeguard against potentially spurious extreme tail behavior while maintaining smoothness of h and its derivative, compare the middle bottom panel of Figure 1 of the main text. Still, linear extrapolation is possible with $\lambda \to \infty$ if desired.

C. Illustration of the random walk prior for δ

In this section, we provide a brief illustration of how the random walk prior for $\boldsymbol{\delta}$ works. For the sake of simplicity, we take a concrete case of $J = 5$. The illustration generalizes easily. In this case, we have $J - 1 = 4$ elements in $\boldsymbol{\delta}$. The first-difference matrix is

$$\mathbf{D} = \begin{bmatrix} -1 & 1 & 0 & 0 \\ 0 & -1 & 1 & 0 \\ 0 & 0 & -1 & 1 \end{bmatrix},$$

and the prior for $\boldsymbol{\delta}$ can be written in terms of $\mathbf{K} = \mathbf{D}^{\mathsf{T}}\mathbf{D}$:

$$\pi(\boldsymbol{\delta}|\tau_\delta^2) \propto \left(\frac{1}{\tau_\delta^2}\right)^{\mathrm{rk}(\mathbf{K})/2} \exp\left(-\frac{1}{2\tau_\delta^2}\boldsymbol{\delta}^T\mathbf{D}^T\mathbf{D}\boldsymbol{\delta}\right)$$

The product $\mathbf{D}\boldsymbol{\delta}$ means:

$$\mathbf{D}\boldsymbol{\delta} = \begin{bmatrix} \delta_2 - \delta_1 \\ \delta_3 - \delta_2 \\ \delta_4 - \delta_3 \end{bmatrix} = \boldsymbol{\delta}_{[2:4]} - \boldsymbol{\delta}_{[1:3]}.$$

The subscript in the expression $\boldsymbol{\delta}_{[i:j]}$, $i < j$, denotes a partition of $\boldsymbol{\delta}$ consisting of the i^{th} to j^{th} elements of $\boldsymbol{\delta}$, including the i^{th} and j^{th} elements, such that $\boldsymbol{\delta}_{[i:j]}$ has dimension (3×1). We can write

$$\pi(\boldsymbol{\delta}|\tau_\delta^2) \propto \left(\frac{1}{\tau_\delta^2}\right)^{\mathrm{rk}(\mathbf{K})/2} \exp\left(-\frac{1}{2\tau_\delta^2}(\boldsymbol{\delta}_{[2:4]} - \boldsymbol{\delta}_{[1:3]})^T(\boldsymbol{\delta}_{[2:4]} - \boldsymbol{\delta}_{[1:3]})\right),$$

which is proportional to the density of a normal distribution with mean vector $\boldsymbol{\delta}_{[1:3]}$ and covariance matrix $\tau_\delta^2\mathbf{I}$, where $\mathbf{I}$ is the (3×3) identity matrix, for $\boldsymbol{\delta}_{[2:4]}$ and a constant prior for δ_1:

$$\delta_1 \sim \text{const}$$
$$\boldsymbol{\delta}_{[2:4]} \sim \mathcal{N}(\boldsymbol{\delta}_{[1:3]}, \tau_\delta^2\mathbf{I}).$$

D. Details on the scale-dependent prior

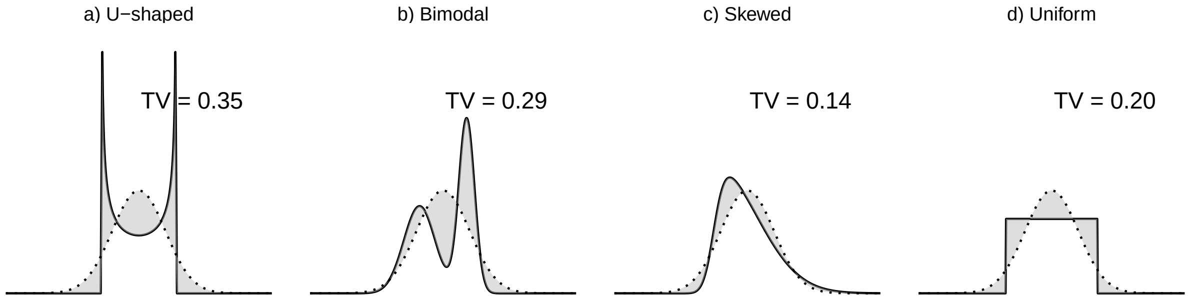


Figure D.1 Illustration of the total variation distance (TV) for four example densities. The total variation distance is the shaded area.

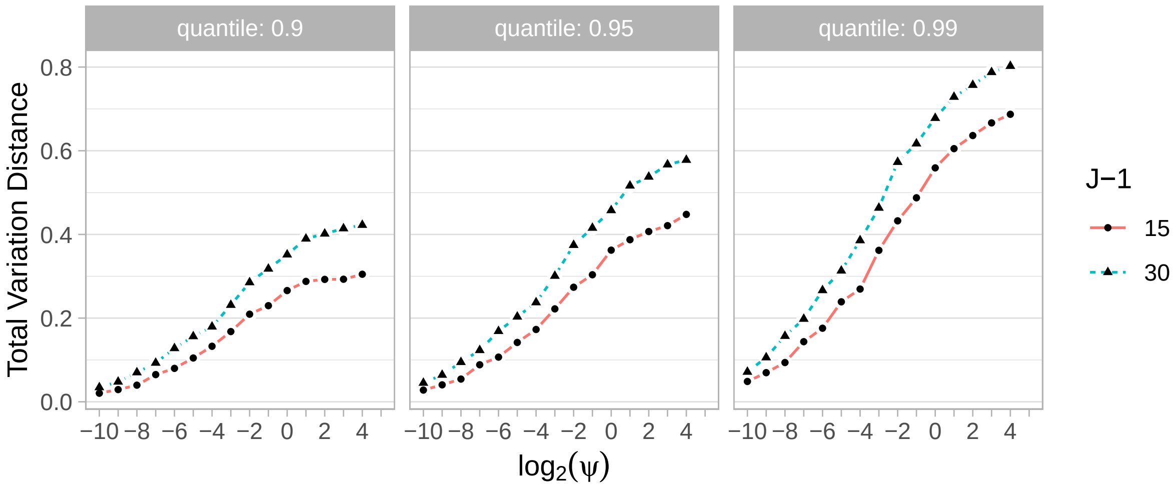


Figure D.2 Quantiles of total variation distances observed using $J-1 = 15$ and $J-1 = 30$.

E. Details on the prior for $\boldsymbol{\delta}$

E.1. Cancellation of constants added to $\boldsymbol{\delta}$ in $h(r)$

First, to provide intuition for the role of δ_1 as an added constant, consider the simple setup where $p(a_1) \propto$ const and $a_2 \sim \mathcal{N}(0, \tau^2)$, collected in $\mathbf{a} = [a_1, a_2]^\top$. Define the shifted vector $\mathbf{c} = \mathbf{a} + b$. Then $c_1 = a_1 + b$ still has a flat prior, since a_1 did, while $c_2 = a_2 + b \sim \mathcal{N}(b, \tau^2)$. Equivalently, we may describe $\mathbf{c}$ in terms of $\mathbf{d} = [d_1, d_2]^\top$, where $d_1 = b$ with $p(d_1) \propto$ const and $d_2 \sim \mathcal{N}(d_1, \tau^2)$. Thus, introducing the constant b is equivalent to reparameterizing the model so that the first component acts as a free baseline, around which the second component is centered. This illustrates how δ_1 plays the role of an added location constant.

Now consider adding a constant $\eta \in \mathbb{R}$ to $\boldsymbol{\delta}$. Then the core segment of the transformation function for $r \in [a, b]$ is

$$\begin{aligned} h(r) &= \alpha + \frac{1}{s(\boldsymbol{\delta} + \eta)} \sum_{j=2}^{J} B_j(r) \sum_{\ell=2}^{j} \exp\big(\delta_{\ell-1} + \eta\big) && (14) \\ &= \alpha + \frac{\exp(\eta)}{s(\boldsymbol{\delta} + \eta)} \sum_{j=2}^{J} B_j(r) \sum_{\ell=2}^{j} \exp\big(\delta_{\ell-1}\big). && (15) \end{aligned}$$

and the normalization to slope one then is

$$\begin{aligned} s(\boldsymbol{\delta} + \eta) &= \frac{1}{b-a} \sum_{j=1}^{J-3} \left(\frac{\exp(\delta_j + \eta) + \exp(\delta_{j+2} + \eta)}{6} + \frac{2\exp(\delta_{j+1} + \eta)}{3} \right) && (16) \\ &= \frac{\exp(\eta)}{b-a} \sum_{j=1}^{J-3} \left(\frac{\exp(\delta_j) + \exp(\delta_{j+2})}{6} + \frac{2\exp(\delta_{j+1})}{3} \right) && (17) \\ &= \exp(\eta) s(\boldsymbol{\delta}), && (18) \end{aligned}$$

such that we have

$$h(r) = \alpha + \frac{\exp(\eta)}{\exp(\eta) s(\boldsymbol{\delta})} \sum_{j=2}^{J} B_j(r) \sum_{\ell=2}^{j} \exp\big(\delta_{\ell-1}\big), \tag{19}$$

where $\exp(\eta)$ cancels out in the fraction such that we have $h(r|\boldsymbol{\delta}) = h(r|\boldsymbol{\delta} + \eta)$ for any $\eta \in \mathbb{R}$.

E.2. Reparameterization of $\boldsymbol{\delta}$

We implement the removal of the unpenalized constant from $\boldsymbol{\delta}$ via reparameterization using a mixed model decomposition. The description here is closely based on the description in Kneib et al. (2019), Appendix A.

1. Find the eigendecomposition of $\mathbf{K} = \boldsymbol{\Gamma}\boldsymbol{\Omega}\boldsymbol{\Gamma}^\top$, where $\boldsymbol{\Omega} = \mathrm{diag}(\omega_1, \ldots, \omega_{J-1})$ with sorted eigenvalues $\omega_1 \leq \cdots \leq \omega_{J-1}$, and $\boldsymbol{\Gamma}\boldsymbol{\Gamma}^\top = \mathbf{I}$ is the orthonormal matrix of corresponding eigenvectors.

2. As the rank of $\mathbf{K}$ is $J-2$, we have $\omega_1 = 0$.

3. We split $\mathbf{\Gamma} = [\mathbf{\Gamma}_1, \mathbf{\Gamma}_2]$ and $\mathbf{\Omega} = \text{blockdiag}(\mathbf{\Omega}_1, \mathbf{\Omega}_2)$, where $\mathbf{\Omega}_1 = \omega_1$ and $\mathbf{\Omega}_2 = \text{diag}(\omega_2, \dots, \omega_{J-1})$. The matrices $\mathbf{\Gamma}_1$ and $\mathbf{\Gamma}_2$ are the corresponding matrices of eigenvectors of dimension $(J-1) \times 1$ and $(J-1) \times (J-2)$.

4. We can now write
$$\boldsymbol{\delta} = \mathbf{\Gamma}_1 \eta + \mathbf{\Gamma}_2 \mathbf{\Omega}_2^{-1/2} \tilde{\boldsymbol{\delta}},$$
with priors
$$\pi(\eta) \propto \text{const} \quad \text{and} \quad \tilde{\boldsymbol{\delta}} \sim \mathcal{N}(\mathbf{0}, \tau_\delta^2 \mathbf{I}_{J-2}).$$
The parameter η represents the part of $\boldsymbol{\delta}$ that is unpenalized by the prior. In mixed model terminology, η corresponds to the fixed effect, while $\tilde{\boldsymbol{\delta}}$ corresponds to the random effect.

5. We can now define $\boldsymbol{\delta} \equiv \mathbf{\Gamma}_2 \mathbf{\Omega}_2^{-1/2} \tilde{\boldsymbol{\delta}}$, with the reduced precision matrix $\mathbf{K} = \mathbf{\Gamma}_2 \mathbf{\Omega}_2 \mathbf{\Gamma}_2^\top$. By this step, the unpenalized constant part of the prior is fixed to zero.

6. We conduct inference on the level of $\tilde{\boldsymbol{\delta}}$.

F. Details on posterior inference

F.1. Basis matrix approximation

To avoid having to compute the basis function evaluations of the transformation function's spline segment in every MCMC iteration, we use a grid approximation. That is, we obtain one matrix of basis function evaluations on a tight equidistant grid $\tilde{r}_1, \dots, \tilde{r}_{1000}$, with $\tilde{r}_1 = a$ and $\tilde{r}_{1000} = b$ set to the lower and upper boundary knot, respectively. For a basis function evaluation of a specific value r, we then find the greatest $\tilde{r}_{\mathrm{lo}}$ and the smallest $\tilde{r}_{\mathrm{hi}}$ such that $r \in [\tilde{r}_{\mathrm{lo}}, \tilde{r}_{\mathrm{hi}}]$ and interpolate between the two corresponding elements $B(\tilde{r}_{\mathrm{lo}})$ and $B(\tilde{r}_{\mathrm{hi}})$ of the pre-computed basis matrix:

$$B_j(r) \approx \left(1 - \frac{r - \tilde{r}_{\mathrm{lo}}}{(b-a)/1000}\right) \cdot B(\tilde{r}_{\mathrm{lo}}) + \frac{r - \tilde{r}_{\mathrm{lo}}}{(b-a)/1000} \cdot B(\tilde{r}_{\mathrm{hi}}).$$

F.2. Initialization

Algorithm 1 details the initialization process. The temporary swap of the variance parameters' priors in Step 2 is a protection against non-concavity induced by potential scale-dependent Weibull priors on the variance parameters. Note that the initialization need not be perfect, it mainly serves to provide sensible starting values for subsequent MCMC inference. Therefore, the temporary alteration of the model in Step 2 is negligible for overall inference. Similar justifications apply for two other features of the initialization algorithm:

1. We jointly optimize the parameters $\boldsymbol{\beta}$, $\boldsymbol{\gamma}$ and $\tilde{\boldsymbol{\delta}}$ with their respective hyperparameters instead of optimizing for the hyperparameters using a marginalized likelihood with $\boldsymbol{\beta}$, $\boldsymbol{\gamma}$ and $\tilde{\boldsymbol{\delta}}$ integrated out; a practice that can induce over-smoothing. This point is partly alleviated by our use of a lower bound > 0 for the variance parameters during optimization.

2. We optimize in stages: We first optimize for the parameters in the location and scale model parts while keeping $\tilde{\boldsymbol{\delta}} = \mathbf{0}$ fixed. We then optimize for $\tilde{\boldsymbol{\delta}}$ and τ^2_δ, keeping the other parameters fixed. Essentially, $\boldsymbol{\beta}$, $\boldsymbol{\gamma}$ and their hyperparameters are found under an auxiliary assumption of Gaussianity. This point is partly alleviated by the fact that, due to the model's assumptions, the transformation function is independent of the location and scale model parts.

Inverse softclip link function. The link function $g : [\underline{x}, \overline{x}] \to \mathbb{R}$ is specified based on the softplus function $\mathrm{sp} : \mathbb{R} \to (0, \infty)$. The softplus function can be modified to work on inputs with arbitrary lower bounds $\underline{x}$ and upper bounds $\overline{x}$, yielding

$$\mathrm{sp}(x) = \ln(1 + \exp(x)) \qquad \mathbb{R} \to (0, \infty) \tag{20}$$

$$\mathrm{sl}(x) = \ln(1 + \exp(x - \underline{x})) + \underline{x} \qquad \mathbb{R} \to (\underline{x}, \infty) \tag{21}$$

$$\mathrm{su}(x) = -\ln(1 + \exp(x - \overline{x})) + \overline{x} \qquad \mathbb{R} \to (0, \overline{x}). \tag{22}$$

The respective inverses are

$$\mathrm{sp}^{-1}(x) = \ln(\exp(x) - 1) \qquad (0, \infty) \to \mathbb{R} \tag{23}$$

$$\mathrm{sl}^{-1}(x) = \ln(\exp(x - \underline{x}) - 1) + \underline{x} \qquad (\underline{x}, \infty) \to \mathbb{R} \tag{24}$$

$$\mathrm{su}^{-1}(x) = \ln(\exp(\overline{x} - x)) + \overline{x} \qquad (0, \overline{x}) \to \mathbb{R}. \tag{25}$$

We define the link function g and its inverse g^{-1} by composing sl and su and their respective inverses, i.e.,

$$g(x) = (\mathrm{sl}^{-1} \circ \mathrm{su}^{-1})(x) \qquad (\underline{x}, \overline{x}) \to \mathbb{R} \tag{26}$$

$$g^{-1}(x) = (\mathrm{su} \circ \mathrm{sl})(x) \qquad \mathbb{R} \to (\underline{x}, \overline{x}). \tag{27}$$

The function g^{-1} is also known under the name *softclip*, since it provides a continuous map from the real line to a bounded interval.

Algorithm 1: Initialization of Bayesian penalized transformation models.

1. Initialize the parameters in $\boldsymbol{\beta}$, $\boldsymbol{\gamma}$ and $\tilde{\boldsymbol{\delta}}$ to zero. Initialize the hyperparameters in $\boldsymbol{\tau}_\ell^2$ and $\boldsymbol{\tau}_s^2$ to ten and τ_δ^2 to one.
2. Replace the prior distributions for all parameters in $(\boldsymbol{\tau}_\ell^2, \boldsymbol{\tau}_s^2)$ and τ_δ^2 with a uniform prior $\mathcal{U}(0.025, 10\,000)$.
3. Transform the variance parameters in $(\boldsymbol{\tau}_\ell^2, \boldsymbol{\tau}_s^2)$ and τ_δ^2 to the real line using the inverse softclip link function $g : [\underline{x}, \overline{x}] \to \mathbb{R}$ using $\underline{x} = 0.025$ and $\overline{x} = 10\,000$. The temporary uniform priors are adjusted accordingly using the change of variables theorem. The function g is given in (26).
4. Find the posterior modes for $(\boldsymbol{\beta}, \boldsymbol{\gamma}, g(\boldsymbol{\tau}_\ell^2), g(\boldsymbol{\tau}_s^2))$ via gradient ascent while holding the remaining parameters fixed at their initial values, i.e.,
$$(\boldsymbol{\beta}^{[0]}, \boldsymbol{\gamma}^{[0]}, g(\boldsymbol{\tau}_\ell^2)^{[0]}, g(\boldsymbol{\tau}_s^2)^{[0]}) = \underset{(\boldsymbol{\beta}, \boldsymbol{\gamma}, g(\boldsymbol{\tau}_\ell^2), g(\boldsymbol{\tau}_s^2))}{\arg\max} \ln p(\boldsymbol{\beta}, \boldsymbol{\gamma}, g(\boldsymbol{\tau}_\ell^2), g(\boldsymbol{\tau}_s^2) | \tilde{\boldsymbol{\delta}}, \tau_\delta^2, \boldsymbol{y}).$$
5. Find the posterior modes for $\tilde{\boldsymbol{\delta}}, g(\tau_\delta^2)$ via gradient ascent while holding the remaining parameters fixed at their values found in the previous step, i.e.,
$$\tilde{\boldsymbol{\delta}}^{[0]}, g(\tau_\delta^2)^{[0]} = \underset{\tilde{\boldsymbol{\delta}}, g(\tau_\delta^2)}{\arg\max} \ln p(\tilde{\boldsymbol{\delta}}, g(\tau_\delta^2), \boldsymbol{y} | \boldsymbol{\beta}^{[0]}, \boldsymbol{\gamma}^{[0]}, g(\boldsymbol{\tau}_\ell^2)^{[0]}, g(\boldsymbol{\tau}_s^2)^{[0]}).$$
6. Transform the variance parameters in $(\boldsymbol{\tau}_\ell^2, \boldsymbol{\tau}_s^2)$ and τ_δ^2 to their original scale using the inverse link function $g^{-1} : \mathbb{R} \to [\underline{x}, \overline{x}]$.
7. Reset the prior distributions for the parameters in $(\boldsymbol{\tau}_\ell^2, \boldsymbol{\tau}_s^2)$ and τ_δ^2 to their original specifications.

F.3. Constant intercepts

We treat the intercepts β_0, γ_0 in the covariate models as constants that are identified by the model assumptions when the remaining parameters are held fixed. We assume $\mathbb{E}(\varepsilon) = 0$ and $\mathbb{V}\mathrm{ar}(\varepsilon) = 1$. To identify β_0 and γ_0, we first write our covariate models for the location and scale without the respective intercept terms, i.e. $\tilde{\mu}(\boldsymbol{x}) = \mu(\boldsymbol{x}) - \beta_0$ and $\tilde{\sigma}(\boldsymbol{x}) = \sigma(\boldsymbol{x}) / \exp(\gamma_0)$. Now, if we treat β_0 and γ_0 as constants, we obtain the expressions

$$\beta_0 = \mathbb{E}\left(\frac{1}{\tilde{\sigma}(\boldsymbol{x})}\right)^{-1} \mathbb{E}\left(\frac{y - \tilde{\mu}(\boldsymbol{x})}{\tilde{\sigma}(\boldsymbol{x})}\right) \quad \text{and} \quad \gamma_0 = \ln\left(\sqrt{\mathbb{V}\mathrm{ar}\left(\frac{y - \beta_0 - \tilde{\mu}(\boldsymbol{x})}{\tilde{\sigma}(\boldsymbol{x})}\right)}\right). \tag{28}$$

During MCMC sampling, we replace the expected value and the variance with their respective sample estimators, so that β_0 and γ_0 are always available as deterministic functions of the data and the remaining model parameters. Consequently, β_0 and γ_0 can be updated via (28) in each iteration of the sampling algorithm.

F.4. IWLS proposals for $\boldsymbol{\beta}_\ell$ and $\boldsymbol{\gamma}_s$

Klein et al. (2015) found that using the expectation of the negative second derivative $\mathbb{E}(\mathbf{F}(\boldsymbol{\theta}^{[t-1]}))$ in $\mathbf{F}(\boldsymbol{\theta}^{[t-1]})^{-1}$ improved sampling efficiency in distributional regression models. For $\boldsymbol{\theta} = \boldsymbol{\beta}_\ell$ and $\boldsymbol{\theta} = \boldsymbol{\gamma}_s$, we can approximate $\mathbb{E}(\mathbf{F}(\cdot))$ by the expected negative second derivatives of the log full conditionals derived under the assumption of a Gaussian likelihood. They are then given by $\mathbb{E}(\mathbf{F}(\boldsymbol{\theta})) \approx \mathbf{B}_\theta^\top \mathbf{W}_\theta \mathbf{B}_\theta + \frac{1}{\tau_\theta^2}\mathbf{K}_\theta$. The matrix $\mathbf{B}_\theta$ is the respective $N \times D$ basis matrix of each term, $\mathbf{K}_\theta$ is the $D \times D$ penalty matrix and τ_θ^2 is the variance parameter corresponding to each term, with N denoting the number of response observations. The matrices $\mathbf{W}_\theta$ are $N \times N$ diagonal weight matrices with elements $w_{ii}^\beta = 1/\sigma(\boldsymbol{x}_i)^2$ and $w_{ii}^\gamma = 2$. Different from the approach described in the main text, this alternative does not rely on initial values $\boldsymbol{\theta}^{[0]}$.

F.5. Clipping of τ_δ^2

We encountered numerical issues in some cases, when the model estimates τ_δ^2 to be very close to zero: Sampling for $\tilde{\boldsymbol{\delta}}$ stopped completely, with no proposals getting accepted. The problem can be solved by putting a lower bound on τ_δ^2 for estimation. Specifically, we bound $\ln \tau_\delta^2$ to be ≥ -11 during sampling, such that $\tau_\delta^2 \geq \exp(-11) \approx 0.000017$.

G. Data-generating mechanisms for simulation studies

The data-generating mechanisms used in our simulations are as follows:

- Gaussian: Draw $r_i \sim \mathcal{N}(0,1)$ for $i = 1, \ldots, N$.

- PTM: Draw $z_i \sim \mathcal{N}(0,1)$ for $i = 1, \ldots, N$. Next, draw $\boldsymbol{\delta}$ from its prior using $\tau_\delta = 0.2$. This is done by drawing $\tilde{\boldsymbol{\delta}} \sim \mathcal{N}(\mathbf{0}, \tau^2 \mathbf{I}_{J-2})$ and computing $\boldsymbol{\delta} = \boldsymbol{\Gamma}_2 \boldsymbol{\Omega}_2^{-1/2} \tilde{\boldsymbol{\delta}}$. See Subsection E.2 for details on $\boldsymbol{\Gamma}_2$ and $\boldsymbol{\Omega}_2$. Next, compute the expectation and variance of the implied distribution with CDF $\tilde{F}_R(r) = \Phi(h(r|\boldsymbol{\delta}))$ by numerical integration; denote them as m and s^2. Compute $\tilde{z}_i = s z_i + m$. Then, numerically invert the transformation function to obtain $r_i = h^{-1}(\tilde{z}_i|\boldsymbol{\delta})$, which then has expectation zero and variance one. We use $J - 1 = 15$ parameters and $-a = b = 3$ and set $\lambda = 0.1(b - a) = 0.6$.

- Skewnorm: Draw z_i for $i = 1, \ldots, N$ from a standardized skew-normal distribution with density $f(z_i) = 2\phi(z_i)\Phi(\alpha z_i)$, where $\phi(\cdot)$ and $\Phi(\cdot)$ denote the standard normal density and cumulative distribution function, respectively and α is the slant parameter. We use $\alpha = 5$ for a notable right-skew. Compute standardized samples $r_i = (z_i - \mathrm{E})/\sqrt{\mathrm{V}}$, where the expectation is $\mathrm{E} = (\alpha/\sqrt{1+\alpha^2}) \cdot \sqrt{2/\pi} \approx 0.78$ and the variance is $\mathrm{V} = 1 - (2/\pi) \cdot (\alpha^2/(1+\alpha^2)) \approx 0.39$.

- Mixture: Draw z_i for $i = 1, \ldots, N$ from a two-component mixture of Gaussian distributions with means $\boldsymbol{\mu} = (-2, 1)$, standard deviations $\boldsymbol{\sigma} = (1, 0.5)$ and mixing probabilities $p_1 = p_2 = 0.5$. This creates a bimodal distribution. Compute standardized samples $r_i = (z_i - \mathrm{E})/\sqrt{\mathrm{V}}$, where the expectation is $\mathrm{E} = p_1\mu_1 + p_2\mu_2 = 0.75$ and the variance is $\mathrm{V} = p_1\sigma_1^2 + p_2\sigma_2^2 + p_1p_2(\mu_1 - \mu_2)^2 = 2.875$.

- U-shaped: Draw $z_i \sim \mathrm{Beta}(a, b)$ for $i = 1, \ldots, N$ with $a = b = 0.5$, which is a U-shaped distribution. Compute standardized samples $r_i = (z_i - \mathrm{E})/\sqrt{\mathrm{V}}$, where the expectation is $\mathrm{E} = a/(a+b) = 0.5$ and the variance is $\mathrm{V} = (ab)/((a+b)^2(a+b+1)) = 0.125$.

- Uniform: Draw $z_i \sim \mathrm{Unif}(0, 0.1)$ for $i = 1, \ldots, N$. Compute standardized samples $r_i = (z_i - \mathrm{E})/\sqrt{\mathrm{V}}$, where the expectation is $\mathrm{E} = 0.5$ and the variance is $\mathrm{V} = 1/12$.

Figure D.1 displays the U-shaped, bimodal, skewed, and uniform densities used here.

H. Simulation study 1: Unconditional model

Our first simulation study is designed to explore the influence of hyperparameters and the effect of prior specifications without considering covariates. We conduct the simulation in two parts. First, we focus on hyperparameter settings for the PTM, comparing model performance on Gaussian data and data generated from varying PTMs. Second, we apply the model to four additional, hand-crafted datasets and compare PTM performance to a Gaussian model with constant priors and to kernel density estimations.

We use sample sizes $N_{\text{train}} \in \{25, 50, 100, 500\}$ to cover small-sample situations. We generate data from the Gaussian, PTM, skewnorm, mixture, u-shaped and uniform scenarios described in Appendix G.

To demonstrate independence from starting values, we initialize each element of $\boldsymbol{\delta}$ with a draw from a Unif$(-1, 1)$ distribution. We set the initial $\tau_\delta = 1$.

Note that our model is misspecified for the U-shaped and uniform data-generating mechanisms, since these produce sharply bounded data, while our model assumes that the response can take values on the full real line. This is intentional; it provides a challenging situation for our model. For each data-generating mechanism, we draw 100 datasets of size $N = 2\,000$. We apply the PTM, the Gaussian model, and the kernel density estimation to a subset $r_1, \dots, r_{N_{\text{train}}}$ observations, where $N_{\text{train}} \in \{25, 50, 100, 500\}$ as mentioned above. We use $1\,500$ observations as testing data when evaluating out-of-sample predictive performance. Note that, since we randomly draw $\boldsymbol{\delta}$, this procedure actually creates a different underlying distributional shape for each dataset of the PTM data-generating mechanism. This is intentional, since it creates a wide variety of shapes and thus allows for a broad exploration of model behavior.

H.1. Performance criteria

We consider the following diagnostic criteria:

- Minimum effective samples drawn per minute for any parameter in $\boldsymbol{\delta}$ and for τ^2_δ. We compute the "bulk" and "tail" effective sample sizes separately. This quantifies sampling efficiency and helps identify potential issues in sampling difficult parameters. Separate evaluation of bulk and tail ESS provides insight into information in central vs. extreme parts of the posterior. The quantities are computed using the Python library Arviz (Kumar et al., 2019), version 0.21.0.

- Maximum $\hat{R}$ for any parameter in $\boldsymbol{\delta}$ and for τ^2_δ. The $\hat{R}$ diagnostic assesses convergence across chains; values close to 1 indicate good convergence. Taking the maximum highlights the worst-performing parameter.

- Acceptance probability for proposed draws. A well-tuned sampler typically achieves acceptance rates within a target range (e.g., 0.6–0.9 in HMC), helping to avoid overly small or overly large step sizes.

- Relative frequency of divergent transitions. Divergences indicate regions of high curvature or problematic geometry and are critical warnings in HMC-based methods.

We consider the Kullback-Leibler divergence (KLD) as a metric for predictive performance. Specifically, we compute the divergence from the true log density to the estimated predictive log density on the test observations with indices $i = 501, \dots, N$ as

$$\text{KLD} = \frac{1}{T(N-500)} \sum_{t=1}^{T} \sum_{i=501}^{N} (\ln f(r_i) - \ln \hat{f}^{(t)}(r_i)).$$

The index $t = 1, \ldots, T$ identifies the individual MCMC samples. Smaller KLD values indicate a better approximation of the true distribution, with a minimum of zero.

H.2. Hyperparameter configurations

We compare seven hyperparameter configurations:

- `base`: A penalized transformation model without covariates, such that $F(r_i) = \Phi(h((r_i - \mu)/\sigma|\boldsymbol{\delta}))$. The prior for $\boldsymbol{\delta}$ is the random walk prior as described in the main text, with random walk variance τ_δ^2. The prior for τ_δ^2 is the scale-dependent prior Weibull$(0.5, \theta)$ with $\theta = 0.5$. We set $-a = b = 4$ and use $J - 1 = 15$ parameters in $\boldsymbol{\delta}$. We set $\lambda = 0.1(b - a)$. We sample $\boldsymbol{\delta}$ and $\log(\tau^2)$ jointly using the No-U-Turn Sampler (NUTS) and use a target acceptance probability of .8 in the warmup for NUTS. We use 5 000 warmup samples and 10 000 posterior samples, each in four independent chains, resulting in a total of 40 000 posterior samples. We consider this configuration as the base configuration. The other configurations are created by varying individual settings.
- `IG`: Like `base`, but using an inverse gamma hyperprior for τ^2, with concentration $a = 0.01$ and scale $b = 0.01$, which are common default settings for producing an uninformative prior. This hyperprior allows us to sample τ_δ^2 using a Gibbs update. The full conditional is $\tau^2|\boldsymbol{\delta} \sim \mathcal{IG}(\tilde{a}, \tilde{b})$ with $\tilde{a} = a + \frac{1}{2}\text{rank}(\mathbf{K})$ and $\tilde{b} = b + \frac{1}{2}\boldsymbol{\delta}^\mathsf{T}\mathbf{K}\boldsymbol{\delta}$, where $\mathbf{K}$ is the penalty matrix used in the prior for $\boldsymbol{\delta}$. In this case, we sample $\boldsymbol{\delta}$ and τ^2 in separate blocks: 1) Draw a sample of $\boldsymbol{\delta}|\tau^2$ using NUTS, 2) draw a sample of τ^2 from its full conditional. The inverse gamma prior is an important comparison, because it is widely used for P-splines and computationally convenient.
- `ap90`: Like `base`, but using a target acceptance probability of .9 in the warmup for NUTS. This leads to smaller step sizes in proposal generation and may make it easier for the model to explore complex posterior landscapes.
- `J30`: Like `base`, but using $J - 1 = 30$ parameters in $\boldsymbol{\delta}$. This equips the model with additional flexibility.
- `scale25`: Like `base`, but using a high scale $\theta = 25$ in the Weibull hyperprior for τ_δ^2.
- `identity_extrap`: Like `base`, but using $\lambda \to 0$ for extrapolating the transformation function with the identity function outside $[a, b]$.
- `linear_extrap`: Like `base`, but using $\lambda \to \infty$ for linear extrapolation of the transformation function.

Figure H.1 summarizes this first part of the simulation study with respect to sampling efficiency and convergence. We note the following:

- The inverse gamma hyperprior (`IG`) leads to superior sampling efficiency and convergence, providing consistently higher numbers of effective samples and lower $\hat{R}$ than the other configurations. The advantage of the inverse gamma prior however seems to vanish with a moderately high sample size of $N_{\text{train}} = 500$.

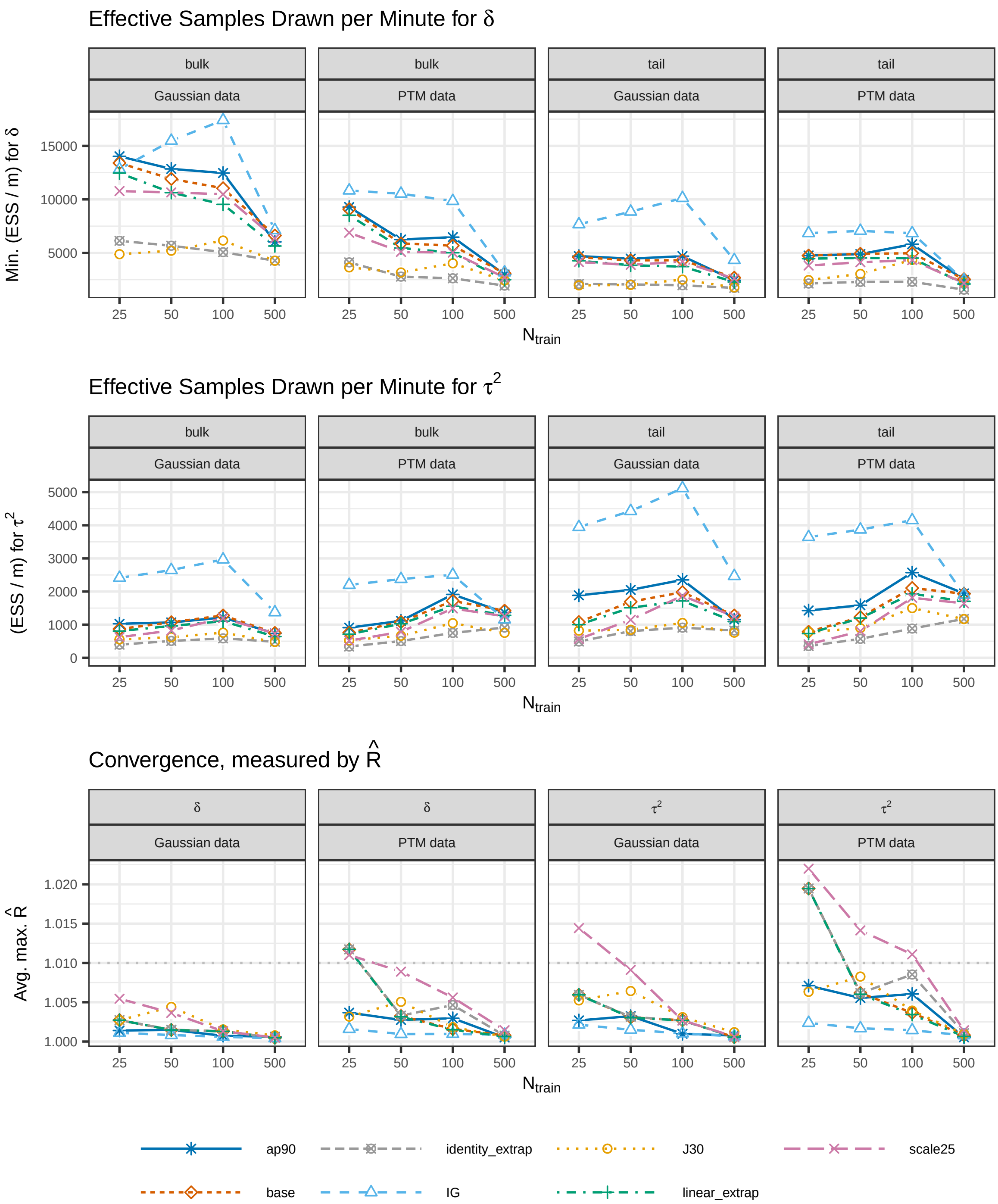


Figure H.1 Effective samples drawn per minute and convergence measure $\hat{R}$ from the first part of the unconditional simulation study, averaged over the results from 100 simulation runs for each sample size, data type, and hyperparameter configuration. For $\boldsymbol{\delta}$, we display the average over the minimum effective sample size for any of the $J-1$ parameters in $\boldsymbol{\delta}$ to provide a conservative estimate.

- With effective samples per minute consistently close to or larger than 2 500, sampling efficiency appears to be generally feasible for moderately-sized datasets.
- Generally, sampling efficiency is higher for $\boldsymbol{\delta}$ than for τ^2 and drops considerably with increasing sample size, indicating that scaling to higher sample sizes is challenging for the PTM.
- Of the configurations with a scale-dependent prior for τ_δ^2, `ap90` appears to be most efficient and show the best convergence diagnostics, with an average $\hat{R} < 1.01$ in all conditions and for all data types.
- The configuration `scale25` appears to converge worse and be less efficient than other configurations.
- Differences between `base`, and `linear_extrap` are small, but `identity_extrap` appears to be notably less efficient with small sample sizes.

Figure H.2 additionally summarizes the achieved acceptance probabilities and divergent transitions that occurred during MCMC sampling. We note the following:

- The acceptance probability is close to the target level in all cases. Generally, acceptance probability rises slightly as sample size increases.
- All configurations show sharply reduced relative frequencies of divergent transitions as sample size increases to $N_{\text{train}} = 500$. For small sample sizes, all configurations except for `ap90` and `IG` show considerable relative frequencies of divergent transitions between one and three percent. `IG` yields the lowest numbers, staying close to 0% divergent transitions at all sample sizes. Of the other configurations, `ap90` shows the lowest relative frequency, staying well below 1% even for the most challenging case of PTM data and $N_{\text{train}} = 25$.

Figure H.3 shows our measures for distributional performance in the first part of the unconditional simulation study. We note the following:

- Performance converges as sample size increases, implying that hyperparameter configurations matter most for small sample sizes.
- `IG` stands out as a configuration that consistently performs slightly worse than the other configurations, particularly for Gaussian data and small sample sizes. This result suggests that the scale-dependent prior for τ_δ^2 better allows the PTM to fall back to the Gaussian base model, if the data do not provide strong evidence for deviations from Gaussianity. Thus, the observation is consistent with the reasoning for our choice of the scale-dependent prior.
- `linear_extrap` performs notably worse than other configurations for PTM data with small sample sizes. Recall that the PTM data were generated using a transition from linear to identity extrapolation, such that the linear extrapolation constitutes a model misspecification in the tails here. It appears reasonable that this pattern would become visible in the Kullback-Leibler divergence. The result suggests that theoretical considerations about the model's desired tail behavior can be relevant if tail behavior is of interest.

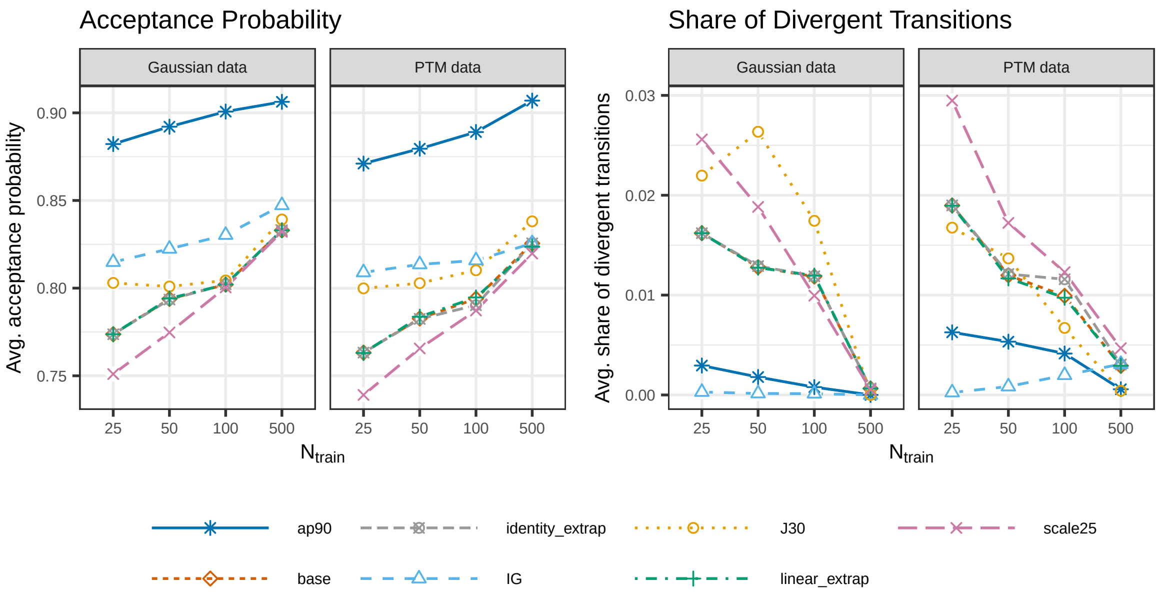


Figure H.2 Acceptance probabilities and relative frequencies of divergent transitions from the first part of the unconditional simulation study, averaged over the results from 100 simulation runs for each sample size, data type, and hyperparameter configuration.

- Except for `linear_extrap` and `IG`, all configurations show practically indistinguishable performance.

Conclusion. In sum, we observe that the scale-dependent prior yields the expected benefit for Gaussian data with small sample sizes. Among the configurations with scale-dependent prior, `ap90` shows superior diagnostic measures in terms of convergence and the relative frequency of divergent transitions, and, to a lesser extent, sampling efficiency, while there is no discernible difference in terms of predictive distributional performance. As a result, we use the configuration `ap90` as the default configuration in the following.

H.3. Different data scenarios

In the second part of the unconditional simulation study, we compare the following models:

- `ptm-15`: A penalized transformation model, using the same configuration as `base` in the first part of the unconditional simulation, but with target acceptance probability .9 for hyperparameter tuning in the warmup period of the No-U-Turn Sampler.
- `ptm-30`: Like `ptm-15`, but using $J - 1 = 30$ instead of $J - 1 = 15$. We include this model, since a higher number of parameters may be useful for the misspecified models.
- `gaussian`: A Gaussian model $r_i \sim \mathcal{N}(\mu, \sigma^2)$, where μ and $\log(\sigma)$ are sampled jointly with the No-U-Turn Sampler using constant priors. Provides a baseline comparison.

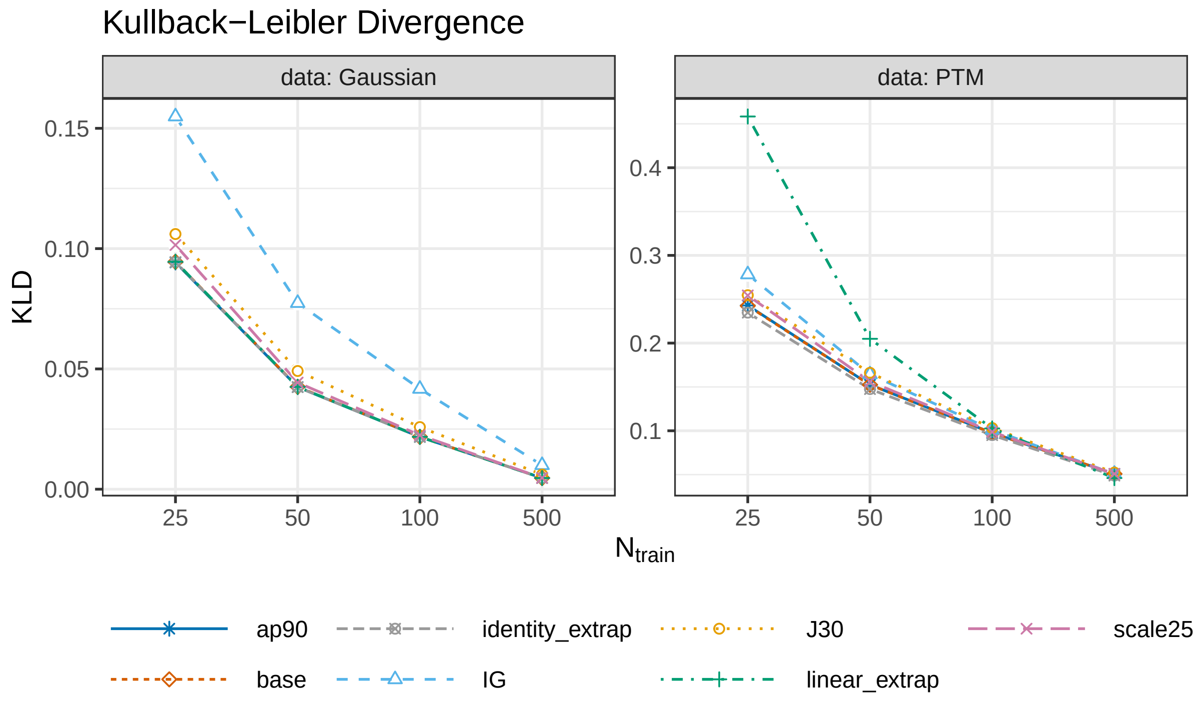


Figure H.3 Kullback-Leibler divergence for evaluating predictive distributional performance in the first part of the unconditional simulation study, averaged over the results from 100 simulation runs for each sample size, data type, and hyperparameter configuration. Smaller values indicate a better performance.

- `kde`: Kernel density estimation using a Gaussian kernel with bandwidth selected via the Sheather–Jones plug-in method (Sheather & Jones, 1991). Provides a baseline comparison for data-driven nonparametric density estimation without uncertainty quantification.

Diagnostic information for the models `ptm-15` and `ptm-30` is displayed in Figure H.4, showing the number of effective samples drawn per minute and convergence measured by $\hat{R}$, and Figure H.5, showing the acceptance probability for MCMC proposals and the relative frequency of divergent transitions. The challenging nature of the investigated scenarios, particularly `Mixture` and the misspecified scenarios `Unif` and `U-shaped` surfaces in a mild tendency for lower sampling efficiency and larger $\hat{R}$, which particularly with small sample sizes crosses the rule-of-thumb threshold of 1.01 in some cases. Nonetheless, $\hat{R}$ stays below 1.02 in all but one case and only minimally above 1.02 in the exception, indicating that convergence issues do not tend to be severe. The acceptance probability of MCMC proposals stays well above 80% in all cases. The relative frequency of divergent transitions remains below 0.02 in all cases and below 0.01 for `ptm-30`. We interpret this as a slight indication in favor of the more complex model `ptm-30`.

Figure H.6 shows the Kullback-Leibler divergence to the true distribution, our metric for evaluating distributional fit. We note the following:

- For Gaussian data, the performance of all models is very similar. Particularly, `ptm-15` and `gaussian` appear to be indistinguishable, while `ptm-30` shows a minimally higher

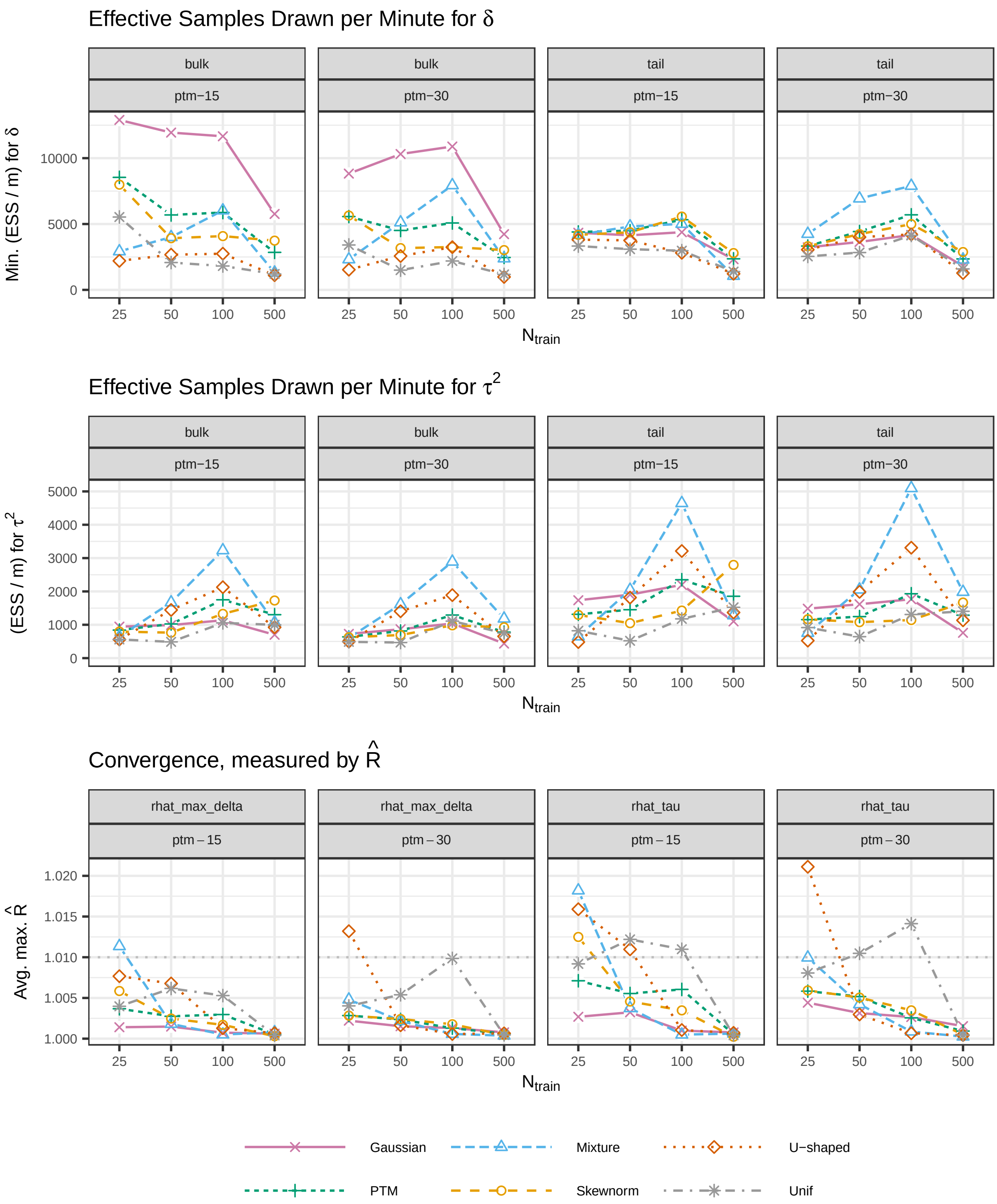


Figure H.4 Effective samples drawn per minute and convergence measure $\hat{R}$ from the second part of the unconditional simulation study, averaged over the results from 100 simulation runs for each sample size, data type, and model. For $\boldsymbol{\delta}$, we display the average over the minimum effective sample size for any of the $J-1$ parameters in $\boldsymbol{\delta}$ to provide a conservative estimate.

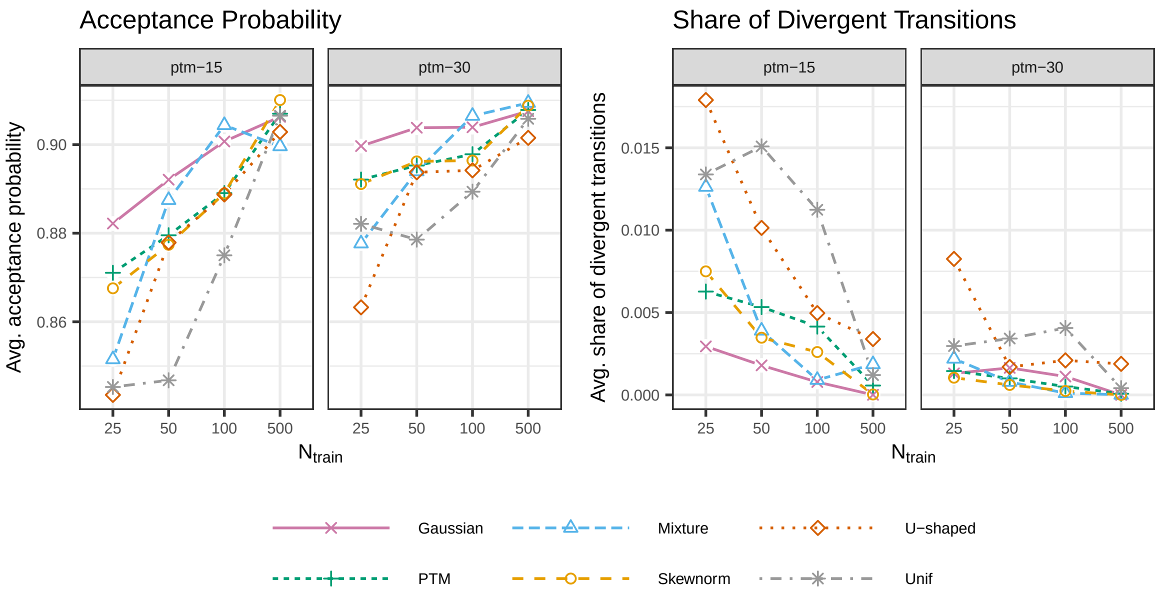


Figure H.5 Acceptance probabilities and relative frequencies of divergent transitions from the second part of the unconditional simulation study, averaged over the results from 100 simulation runs for each sample size, data type, and model.

divergence with small sample sizes. The kernel density estimate again shows a minimally higher divergence than `ptm-30`.

- In all scenarios, the performance of the PTMs converges to the performance of the kernel density estimation with growing sample size. With $N_{\text{train}} = 100$, the difference to kernel density estimation can be considered small, and with the next higher sample size in our setup, $N_{\text{train}} = 500$, PTM performance is almost indistinguishable from kernel density estimation in all scenarios.
- For `PTM` data and `Skewnorm` data, the patterns look very similar. All models perform roughly on the same level with the smallest sample size $N_{\text{train}} = 25$, and differences start to show and grow larger as sample size increases. Specifically, the divergence measured for `ptm-15`, `ptm-30` and `kde` drops notably faster than it does for `gaussian`; and the three former models appear to get closer to the true distribution at a similar pace.
- For `Mixture` data and `Unif` data, the divergence measured for `ptm-15`, `ptm-30` and `kde` again drops notably faster than it does for `gaussian`. However, `kde` gets closer to the true distribution than the PTM models with small sample sizes, the performance only converges with $N_{\text{train}} = 500$.
- For `U-shaped` data, `kde` likewise gets closer to the true distribution than the PTMs with small sample sizes. However, `ptm-30` performs better than `ptm-15`, getting close to `kde` with $N_{\text{train}} = 50$ and becoming indistinguishable from it at $N_{\text{train}} = 100$.

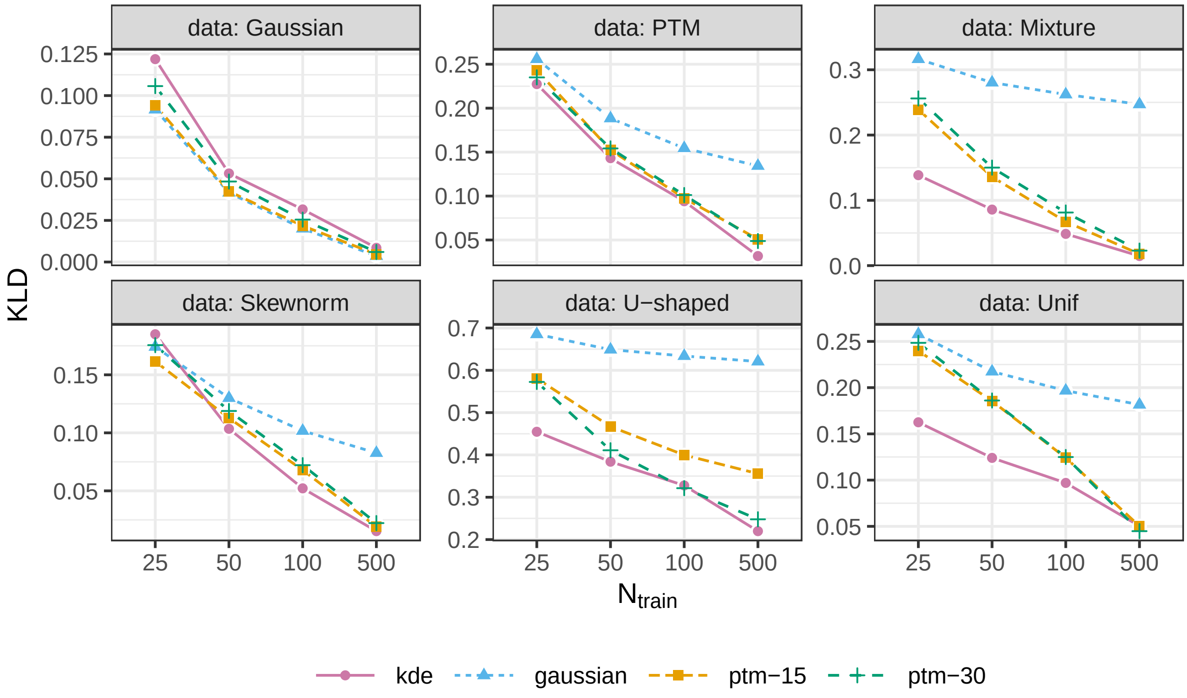


Figure H.6 Kullback-Leibler divergence for evaluating predictive distributional performance in the second part of the unconditional simulation study, averaged over the results from 100 simulation runs for each sample size, data type, and model. Smaller values indicate a better performance.

In this part of the simulation, we also considered an aggregated estimate of pointwise calibration of highest posterior density intervals for the CDF, computed as

$$\text{Coverage}(\hat{F}) = \frac{1}{N-500} \sum_{i=501}^{N} \mathbb{I}(F_R(r_i) \in [\hat{F}_{\text{low}}(r_i), \hat{F}_{\text{high}}(r_i)]),$$

where $\hat{F}_{\text{low}}(r_i)$ and $\hat{F}_{\text{high}}(r_i)$ are the lower and upper boundaries of a 0.90 highest posterior density interval computed on the full sample of posterior CDF evaluations $\hat{F}^{(t)}(r_i), t = 1, \ldots, T$ and $\mathbb{I}(\cdot)$ denotes the indicator function for the enclosed condition. Figure H.7 shows the results. We can observe that the PTMs yield highest posterior density intervals with close to nominal coverage, and that coverage converges to the nominal level as sample size increases. Unsurprisingly, the same is not true for the Gaussian baseline comparison model, which yields similar coverage levels as the PTMs only for Gaussian data and the smallest sample size $N_{\text{train}=25}$. In the `U-shaped` data scenario, the `ptm-15` model shows a notable drop in coverage as sample size increases from $N_{\text{train}=100}$ to $N_{\text{train}=500}$, arriving at .615 for $N_{\text{train}=500}$. While this is still drastically higher than the Gaussian comparison at .158, it may reflect a limitation of the less flexible model with $J - 1 = 15$ parameters in the transformation function to accurately capture the features of the observed data; thus growing more certain about a suboptimal fit. While the more flexible `ptm-30` at .797 also does not reach the nominal level, it stays much closer, indicating that the added flexibility helps.

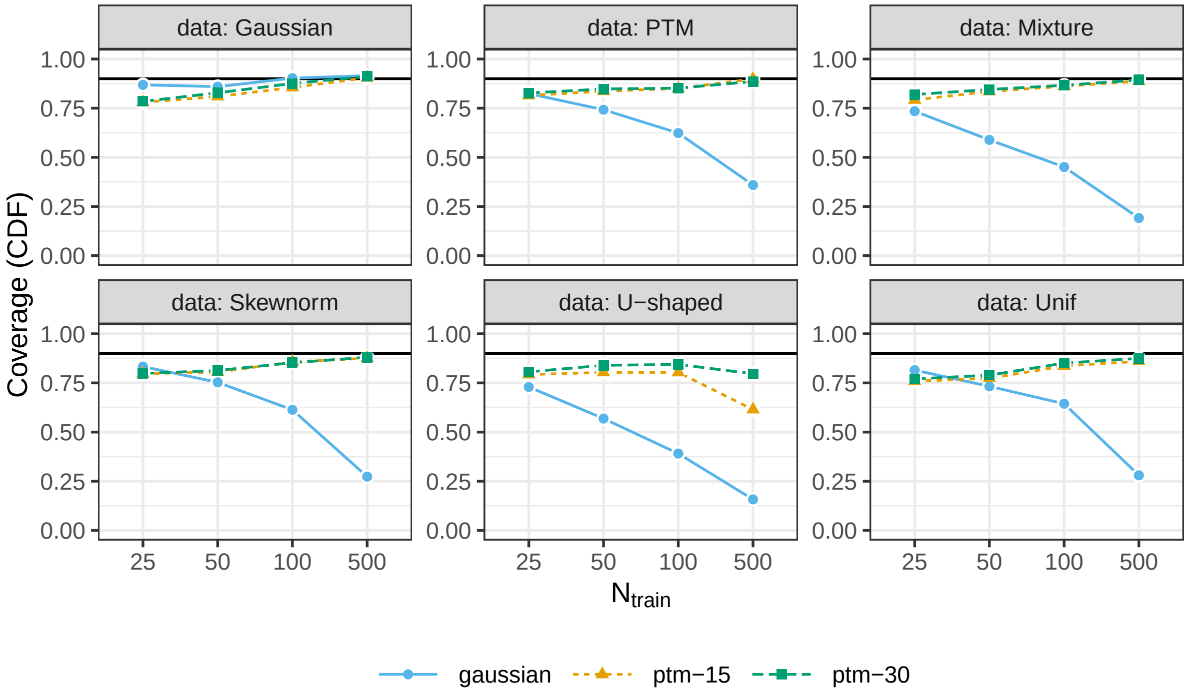


Figure H.7 Average pointwise coverage probability for 90% highest posterior density intervals. The target level .90 is marked by a solid black horizontal line.

Conclusion. As the PTMs perform better than the Gaussian distribution in all non-Gaussian scenarios and converge to be on par with kernel density estimation as sample size grows, we interpret these results as demonstrating the ability of PTMs to effectively capture distributional information from observed data. Notably, the convergence of PTM and kernel density estimation performance even holds for challenging scenarios like the multi-modal `Mixture` case and, due to their strict boundedness, drastically misspecified scenarios like `U-shaped` and `Unif`. The discrepancy to kernel density estimation at smaller sample sizes in these three scenarios can be attributed to our prior setup in the PTMs, which penalizes the model towards the Gaussian reference distribution. For PTMs, strong deviations from this reference are only warranted with strong empirical evidence, so the observed pattern shows us that the model is behaving as expected and intended. We observe some indications that a higher number of parameters in the transformation function provides benefits without doing harm: First, we observe a smaller relative frequency of divergent transitions with `ptm-30` than with `ptm-15`; second, we observe superior performance of `ptm-30` compared to `ptm-15` in terms of Kullback-Leibler divergence and coverage; and third, we do not observe substantial disadvantages of `ptm-30` compared to `ptm-15` in terms of any other diagnostic or performance measure.

I. Simulation study 2: Conditional model

I.1. Covariate functions

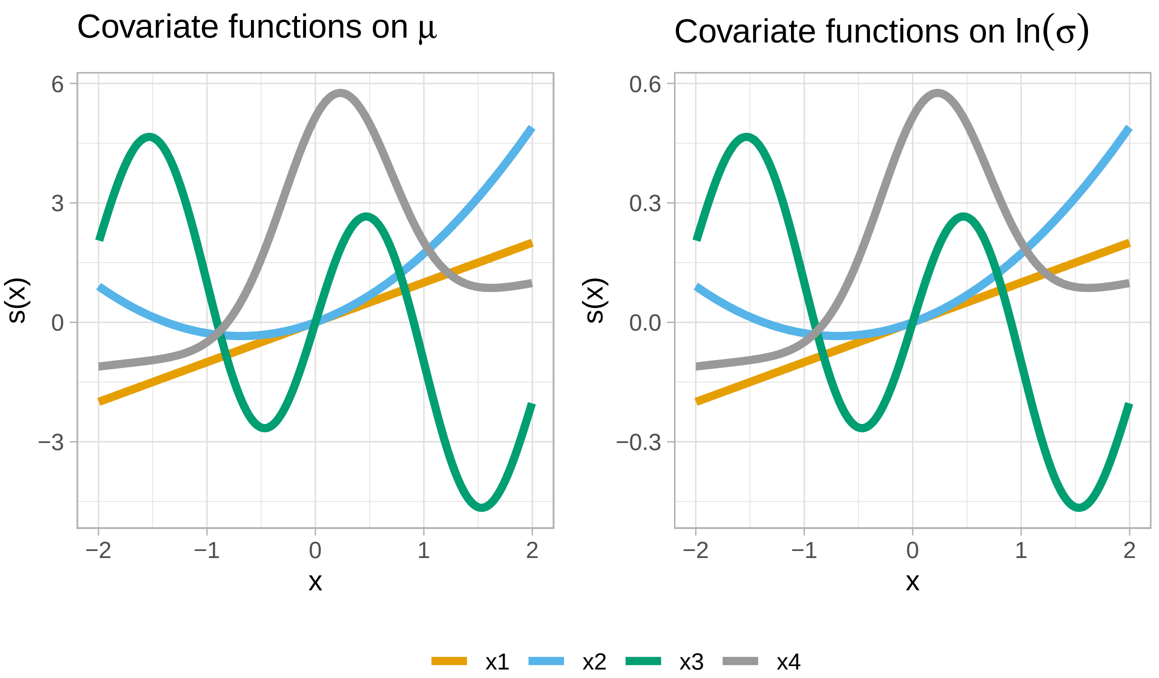


Figure I.1 Covariate functions used in the simulation study. Note the different scales of the y-axes. The lower scale for the scale functions is due to them operating on log-level, leading to multiplicative effects on the level of σ.

I.2. Model specifications

- PTM: Bayesian penalized transformation models (PTMs) with a standard Gaussian reference distribution and $J - 1 = 30$ transformation function parameters in $\boldsymbol{\delta}$. The prior for $\boldsymbol{\delta}$ is the random walk prior described in Subsection 2.2, with random walk variance $\tau_\delta^2 \sim \text{Weibull}(0.5, 0.5)$. We set up the knot base with $-a = b = 4$, and use $\lambda = 0.1(b - a)$ in the transformation function. The models can be written as

$$\mathbb{P}(Y \leq y_i) = \Phi\left(h\left(\frac{y_i - \mu(\boldsymbol{x}_i)}{\sigma(\boldsymbol{x}_i)}\right)\right),$$

where $\boldsymbol{x}_i = [x_{i1}, x_{i2}, x_{i3}, x_{i4}]^\mathsf{T}$. The location model part is specified as

$$\mu(\boldsymbol{x}_i) = \beta_0 + s_1(x_{i1}) + s_2(x_{i2}) + s_3(x_{i3}) + s_4(x_{i4}),$$

where each $s_1, \ldots, s_4$ is a 20-parameter Bayesian P-spline. The inverse smoothing parameters of these P-splines receive inverse gamma hyperpriors with concentration 1 and scale

0.001. The scale model part is specified analogously, additionally using a logarithmic link function. For estimation, we apply MCMC as described in the main text. We draw 25 000 samples each in four chains after a warmup of 5 000 iterations and use a thinning factor of 5, yielding a total of 20 000 samples.

- Gaussian: Gaussian location-scale models, which can be written in analogy to PTMs as

$$\mathbb{P}(Y \leq y_i) = \Phi\left(\frac{y_i - \mu(\boldsymbol{x}_i)}{\sigma(\boldsymbol{x}_i)}\right)$$

by letting the transformation function h equal the identity function. The covariate models $\mu(\boldsymbol{x}_i)$ and $\sigma(\boldsymbol{x}_i)$ are specified exactly as in the PTMs. For estimation, we apply MCMC with iteratively weighted least squares proposals (Klein et al., 2015) in Liesel (Riebl et al., 2023). We draw 50 000 samples each in four chains after a warmup of 5 000 iterations and use a thinning factor of 10, yielding a total of 20 000 samples.

- TAMLS: A transformation additive model (Siegfried et al., 2023) of the form

$$\mathbb{P}(Y \leq y_i) = \Phi\left(\sigma(\boldsymbol{x}_i)^{-1} h(y_i) - \mu(\boldsymbol{x}_i)\right),$$

with

$$\mu(\boldsymbol{x}_i) = \beta_0 + s_1(x_{i1}) + s_2(x_{i2}) + s_3(x_{i3}) + s_4(x_{i4}). \tag{29}$$

where each $s_1, \ldots, s_4$ is a 20-parameter P-spline. The scale model part is specified analogously, but using a logarithmic link function and omitting the intercept for identifiability. Implementation was done by adapting the example code included in the R package `tram`, available on GitHub: https://github.com/cran/tram/blob/master/demo/stram.R.

- SBGP: A semiparametric Bayesian Gaussian process model as proposed by Kowal and Wu (2024). The model is $g(y_i) = f_\theta(\boldsymbol{x}_i) + \sigma\epsilon_i$ where $\boldsymbol{x}_i = (x_{i1}, x_{i2}, x_{i3}, x_{i4})^\mathsf{T}$, $\epsilon_i \sim \mathcal{N}(0,1)$ and $f_\theta \sim \mathcal{GP}(m_\theta, K_\theta)$. In the Gaussian process, m_θ is an unknown constant mean and K_θ is an isotropic Matérn covariance function with unknown variance, range, and smoothness parameters. We tested using an anisotropic Matérn covariance function instead, but encountered overflow errors in this model variant. We also tested a version of the model that includes a simple linear term $g(y_i) = \boldsymbol{x}_i^\mathsf{T}\boldsymbol{\beta} + f_\theta(\boldsymbol{x}_i) + \sigma\epsilon_i$. We found that the results improved when including this term and report the results for this model version. Computations are carried out using the code of the R package `SeBR` as provided by the authors in the supplementary materials of Kowal and Wu (2024). We save 5 000 posterior predictive samples for each row of the test dataset.

- DDPstar: A density regression via Dirichlet process mixtures of normal structured additive regression models (Rodríguez-Álvarez et al., 2024). The model sets up the conditional density of y_i given $\boldsymbol{x}_i = (x_{i1}, x_{i2}, x_{i3}, x_{i4})^\mathsf{T}$ as

$$p(y_i|\boldsymbol{x}_i) = \int \phi(y_i|\mu(\boldsymbol{x}_i), \sigma^2) \mathrm{d}G_{\boldsymbol{x}_i}(\mu(\boldsymbol{x}_i), \sigma^2),$$

with $\phi(\cdot|\mu, \sigma^2)$ denoting the Gaussian density with mean μ and variance σ^2. The means of the mixture components are modeled as

$$\mu_l(\boldsymbol{x}_i) = \beta_{0,l} + s_{1,l}(x_{i1}) + s_{2,l}(x_{i2}) + s_{3,l}(x_{i3}) + s_{4,l}(x_{i4}), \tag{30}$$

where each $s_1, \dots, s_4$ is a 20-parameter P-spline using the default concentration 1 and scale 0.005. As Rodríguez-Álvarez et al. (2024) do in their example code, we use 20 mixture components, and otherwise stick to the defaults of their R library `DDPstar`, which we use to carry out the computations. We draw 15 000 samples after a warmup of 5 000 iterations and use a thinning factor of 10, yielding a total of 1 500 samples. The aggressive thinning was necessary to keep the computation of performance criteria feasible.

- BCTM-LS: Bayesian conditional transformation models (BCTM, Carlan et al., 2023) with standard normal reference distribution. These BCTMs use the form

$$\mathbb{P}(Y \leq y_i) = \Phi(\gamma_0 + h_0(y) + h_1(x_{i1}) + h_2(x_{i2}) + h_3(x_{i3}) + h_4(x_{i4})),$$

where $h_1, \dots, h_4$ are 20-parameter Bayesian P-splines and h_0 is a 20-parameter monotonically increasing Bayesian P-spline. Overall intercepts are removed from the partial transformation functions, and a global intercept is added back into the model as γ_0. We use inverse gamma hyperpriors with concentration 1 and scale 0.001 for all inverse smoothing parameters of the P-spline terms. We draw 5 000 samples each in four chains after a warmup of 5 000 iterations, yielding a total of 20 000 samples. Since sampling is done exclusively with NUTS, we use no thinning.

- BCTM-TE: Bayesian conditional transformation models (BCTM, Carlan et al., 2023) with standard normal reference distribution. These BCTMs use the form

$$\mathbb{P}(Y \leq y_i) = \Phi(h_1(y_i|x_{i1}) + h_2(y_i|x_{i2}) + h_3(y_i|x_{i3}) + h_4(y_i|x_{i4})),$$

where

$$h_j(y_i|x_{ij}) = (\gamma_0 + \boldsymbol{a}(y_i)^\top \otimes \boldsymbol{b}_j(x_{ij})^\top)^\top \boldsymbol{\gamma}_j, \qquad j = 1, \dots, 4,$$

are the BCTM's partial transformation functions, with $\otimes$ denoting the Kronecker product. The bases $\boldsymbol{a}$ and $\boldsymbol{b}$ are chosen as 8-parameter B-spline bases. Overall intercepts are removed from the partial transformation functions, and a global intercept is added back into the model as γ_0. The BCTM uses inverse gamma hyperpriors with concentration 1 and scale 0.001 for all inverse smoothing parameters of the tensor product terms. We apply MCMC via the No-U-Turn Sampler as available in the Python library `liesel-bctm` (Brachem et al., 2023). We draw 5 000 samples each in four chains after a warmup of 5 000 iterations, yielding a total of 20 000 samples. Since sampling is done exclusively with NUTS, we use no thinning.

- QGAM: A fully specified additive quantile regression model with P-splines for all four covariates, as available in the R package `qgam` (Fasiolo et al., 2021a; Fasiolo et al., 2021b). We fit QGAM for a grid of 25 evenly spaced probability levels in $\{0.005, \dots, 0.995\}$. Each quantile model is of the form

$$Q_Y(\alpha|\boldsymbol{x}_i) = \beta_0 + s_{\alpha,1}(x_{i1}) + s_{\alpha,2}(x_{i2}) + s_{\alpha,3}(x_{i3}) + s_{\alpha,4}(x_{i4}),$$

where each $s_{\alpha,1}, \dots, s_{\alpha,4}$ is a 20-parameter P-spline and $Q_Y(\alpha|\boldsymbol{x}_i)$ is the conditional α-quantile of the response Y. We include a model $\kappa(\boldsymbol{x})$ for the variance of the preliminary location-scale GAM fit used in the qgam inference algorithm. The term $\kappa(\boldsymbol{x})$ includes 7-parameter P-splines for all covariates.

I.3. Performance criteria

1. The widely-applicable information criterion (WAIC) can be viewed as an estimator for the expected log pointwise predictive density (Vehtari et al., 2017; Watanabe, 2010). It thus measures predictive performance. On the deviance scale, the WAIC is given by $\text{WAIC} = -2(\text{lppd} - p_{\text{WAIC}})$, where the log pointwise predictive density (lppd) is

$$\text{lppd} = \sum_{i=1}^{N} \ln \left(\frac{1}{T} \sum_{t=1}^{T} \hat{f}_Y^{[t]}(y_i|\boldsymbol{x}_i) \right)$$

and the effective number of parameters is given by

$$p_{\text{WAIC}} = \sum_{i=1}^{N} \widehat{\text{Var}}_{t=1}^{T} \left(\ln \hat{f}_Y^{[t]}(y_i|\boldsymbol{x}_i) \right).$$

The WAIC is a fully Bayesian model choice criterion and can be applied without knowledge of the true distribution. On the deviance scale, a smaller WAIC indicates a better predictive performance. Here, N refers to the size of the training data set.

2. The mean absolute difference (MAD) between the true cumulative density function (CDF) evaluations of the test data and the estimated CDF for the test data, given by

$$\text{MAD}^{[t]} = \frac{1}{N_{\text{test}}} \sum_{i=1}^{N_{\text{test}}} \left| F_{\text{true}}(y_i|\boldsymbol{x}_i) - \hat{F}^{[t]}(y_i|\boldsymbol{x}_i) \right|.$$

3. The Kullback-Leibler divergence (KLD) is a common measure to assess the statistical distance between two distributions. We estimate the KLD as

$$\widehat{\text{KLD}}^{[t]} = \frac{1}{N_{\text{test}}} \sum_{i=1}^{N_{\text{test}}} \left(\ln f_{\text{true}}(y_i|\boldsymbol{x}_i) - \ln \hat{f}^{[t]}(y_i|\boldsymbol{x}_i) \right).$$

4. The continuous ranked probability score (CRPS) is given by

$$\text{CRPS} = \int_0^1 \text{QS}_\alpha(\hat{F}, y)\ \text{d}\alpha, \tag{31}$$

using a generic observations y here, see Gneiting and Raftery (2007) and Gneiting (2011). We use different methods to compute the CRPS, based on availability of computations:

   - Predictive samples. The CRPS can be estimated by comparing a set of predictive samples to a set of test observations. Implementations are available in the R package `scoringRules` (Jordan et al., 2019) and the Python library `properscoring` ("Properscoring: Proper Scoring Rules in Python", 2015). We use this method for SBGP, Gaussian, and PTM.
   - Integrating over the quantile score. We compute the quantile score QS_α for a probability level α for a test observation y_i as

     $$\text{QS}_\alpha(\hat{F}, y_i) = \frac{2}{T}\sum_{t=1}^{T}\Big((\mathbb{I}_{y_i < \hat{F}^{-1,[t]}(\alpha)} - \alpha) \cdot (\hat{F}^{-1,[t]}(\alpha) - y_i) \Big),$$

     where $\mathbb{I}_{(\cdot)}$ is the indicator function for the condition $(\cdot)$. We evaluate the quantile score for a grid of 25 probability levels α, evenly spaced between 0.005 and 0.995, and evaluate the integral in (31) numerically using the trapezoid rule. We use this method for TAMLS, DDPstar, and QGAM.
   - For the BCTM models, we do not record CRPS computations in the simulation study, since we do not have an efficient method for computation available for these models; and they can be compared by KLD and WAIC instead.

5. CDF credible and confidence intervals: For the Bayesian models, we compute credible intervals based on the empirical quantiles of the empirical posterior samples $\{\hat{F}_Y^{[t]}(y_i)\}_{t=1}^T$ for each test observation $i = 1, \ldots, N_{\text{test}}$. We then note whether the true CDF $F_Y(y_i)$ lies inside the so-constructed interval and record a value of 1 if that is the case and 0 otherwise. We compute the coverage by averaging these indicators over all test observations. We compute the width by averaging over the difference of the upper and lower quantile for all test observations. This method is applicable for DDPstar, Gaussian, PTM, BCTM-LS, and BCTM-TE. For TAMLS, we used bootstrapping, then applied the same approach to the bootstrapped samples.
6. MSE of covariate effect: We measure the performance of the covariate effect estimators using the mean squared error, $\text{MSE}(\hat{s}^{[t]}) = \frac{1}{N}\sum_{i=1}^N \big(s_{\text{true}}(x_i) - \hat{s}^{[t]}(x_i)\big)^2$, averaged over all T samples.
7. Credible and confidence intervals for covariate effects: To avoid side-effects from sum-to-zero constraints applied to P-splines, we center all true and estimated covariate functions

within each sample using the test observations. We then apply the same procedure as for the CDF credible and confidence intervals.

Evaluating the conditional CDF and density for SBGP

The semiparametric Bayesian Gaussian Process model (SBGP) implementation provided by Kowal and Wu (2024) currently does not include functionality for directly evaluating the conditional CDF and density. We compute them based on the following considerations using the quantities provided by the implementation. The SBGP model is

$$g(y_i) = f_\theta(\boldsymbol{x}_i) + \sigma\epsilon_i, \quad \epsilon_i \sim \mathcal{N}(0,1)$$

where $i = 1, \ldots, N$ identifies observations. This implies

$$F_Y(y_i \mid \boldsymbol{x}_i, \sigma) = \Phi\left(\frac{g(y_i) - f_\theta(\boldsymbol{x}_i)}{\sigma}\right)$$

where Φ denotes the standard normal CDF. The corresponding conditional density follows by change of variables:

$$p_Y(y_i \mid \boldsymbol{x}_i, \sigma) = \phi\left(\frac{g(y_i) - \mu_i}{\sigma}\right)\frac{1}{\sigma}g'(y_i),$$

where ϕ denotes the standard normal density and g' the derivative of g with respect to y.

Since the SBGP implementation allows access to posterior draws $g^{[t]}$, of the transformation and the scale $\sigma^{[t]}$, $t = 1, \ldots, T$, we get posterior draws of the conditional CDF and PDF, $\hat{F}_Y^{[t]}(y \mid \boldsymbol{x}, \sigma^{[t]})$ and $\hat{p}_Y^{[t]}(y \mid \boldsymbol{x}, \sigma^{[t]})$. These draws use a fixed predictive mean of the latent Gaussian process at $\boldsymbol{x}_i$, as provided by SBGP. For training sample sizes > 1000, SBGP uses a fixed σ at its point estimate instead of samples. We use evaluations of the CDF and PDF on the test data for the Kullback-Leibler divergence and the mean absolute deviation (MAD) of the CDF, and evaluations on the training data for the WAIC.

I.4. Diagnostics

Table I.1 Diagnostic information for PTMs in the simulation study. The model PTM-jitter denotes a PTM with jittered initial values drawn from $\mathcal{N}(0, 1)$ for all parameters. Results are grouped by sample size and MCMC kernel and aggregated over all seeds and data types. The column α displays the average acceptance probability for proposals in the respective kernel. Proposals from Gibbs kernels always get accepted. Proposals from IWLS and NUTS kernels are consistent with the target acceptance probabilities of .5 and .9, respectively. The column Max $\hat{R}$ shows the average maximum $\hat{R}$ within each model run; the maximum is taken over all sampled parameters. The effective sample sizes are average minimums over all parameters' ESS within each model run. The two left ESS columns show ESS per minute.

					Minimum Effective Sample Size			
Model	MCMC	N_{train}	α	Max. $\hat{R}$	Bulk / Min.	Tail / Min.	Bulk	Tail
PTM	GibbsKernel	250	1.00	1.00	419.9	685.9	4209.0	6896.8
PTM	GibbsKernel	500	1.00	1.00	384.2	599.9	5707.0	8948.9
PTM	GibbsKernel	1000	1.00	1.00	276.6	409.4	7258.9	10811.2
PTM	IWLSKernel	250	0.51	1.00	489.9	655.6	4874.0	6524.9
PTM	IWLSKernel	500	0.51	1.00	435.5	567.0	6353.8	8318.5
PTM	IWLSKernel	1000	0.51	1.00	297.9	376.9	7514.6	9673.1
PTM	NUTSKernel	250	0.88	1.01	417.6	421.2	4143.6	4273.6
PTM	NUTSKernel	500	0.89	1.00	448.1	491.4	6764.4	7566.7
PTM	NUTSKernel	1000	0.90	1.00	329.2	369.8	8991.7	10374.9
PTM-jitter	GibbsKernel	250	1.00	2.71	0.4	1.4	5.4	20.9
PTM-jitter	GibbsKernel	500	1.00	2.64	0.4	1.5	5.8	21.8
PTM-jitter	GibbsKernel	1000	1.00	2.55	0.3	1.3	6.6	37.7
PTM-jitter	IWLSKernel	250	0.27	Inf	0.3	0.3	4.3	4.4
PTM-jitter	IWLSKernel	500	0.26	4.84	0.3	0.3	4.5	4.6
PTM-jitter	IWLSKernel	1000	0.31	4.43	0.2	0.2	4.8	5.2
PTM-jitter	NUTSKernel	250	0.83	Inf	0.3	0.5	4.2	7.9
PTM-jitter	NUTSKernel	500	0.80	Inf	0.3	0.4	4.2	7.2
PTM-jitter	NUTSKernel	1000	0.76	Inf	0.2	0.2	4.2	6.5

Table I.2 Notes captured during MCMC sampling for the PTMs in the simulation study. The values are averaged over all seeds, data types, and sample sizes.

Model	Note	Rel. freq.
PTM	divergent transition	0.004
PTM	maximum tree depth	0.000
PTM	nan acceptance prob	0.000
PTM-jitter	divergent transition	0.939
PTM-jitter	maximum tree depth	0.889
PTM-jitter	nan acceptance prob	0.363

I.5. Additional results

Table I.3 Comparison of performance of PTMs and the true Gaussian model on Gaussian data.

Model	N_{train}	KLD ↓	CRPS ↓	WAIC ↓	MAD ↓	CDF CI	
						Coverage	Width
Gaussian	1000	0.028	0.817	3417	0.061	0.923	0.184
PTM	1000	0.028	0.817	3415	0.061	0.923	0.184
Gaussian	500	0.052	0.829	1726	0.084	0.917	0.248
PTM	500	0.052	0.829	1724	0.084	0.916	0.247
Gaussian	250	0.097	0.853	876	0.114	0.910	0.332
PTM	250	0.097	0.853	873	0.114	0.908	0.330

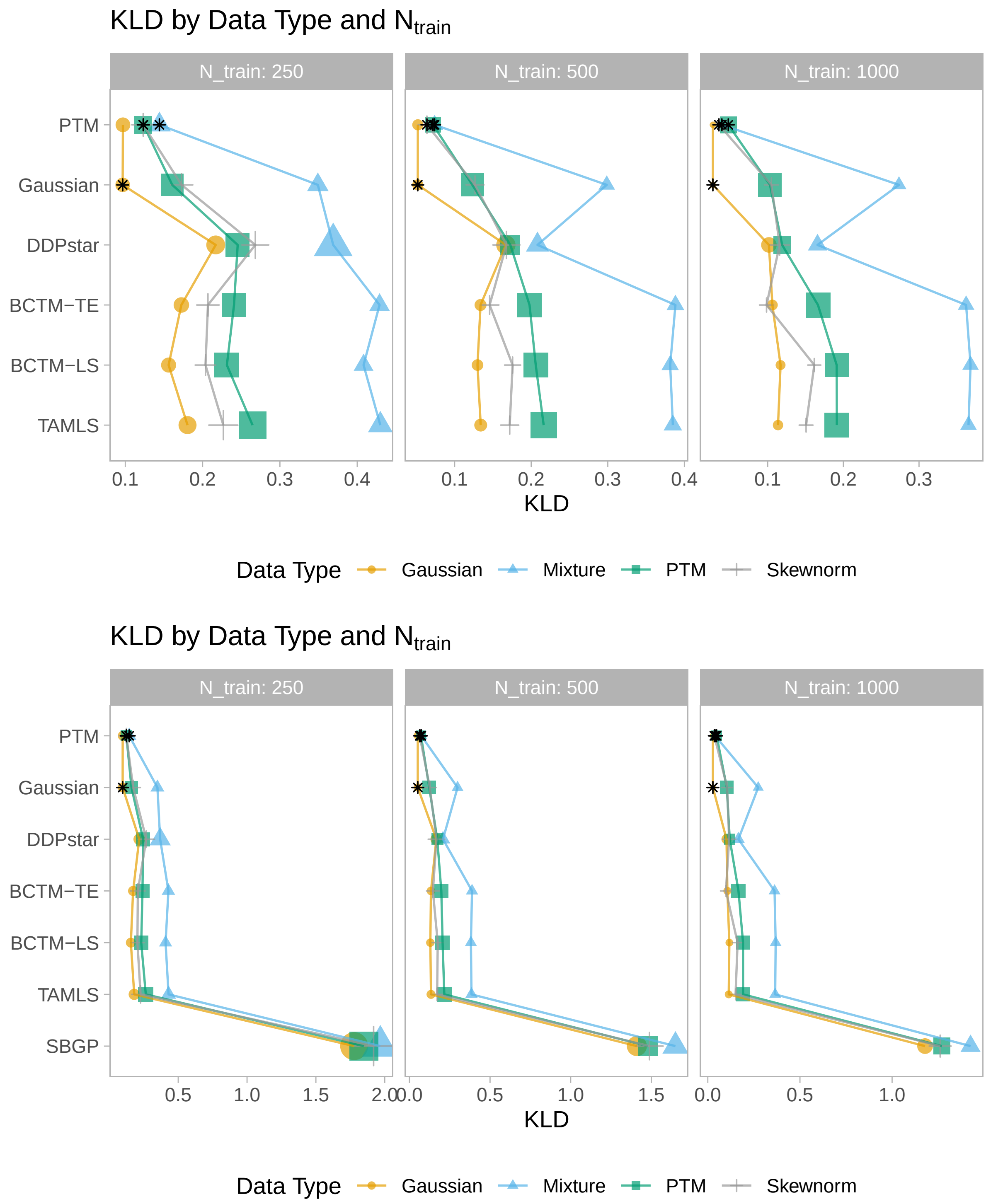
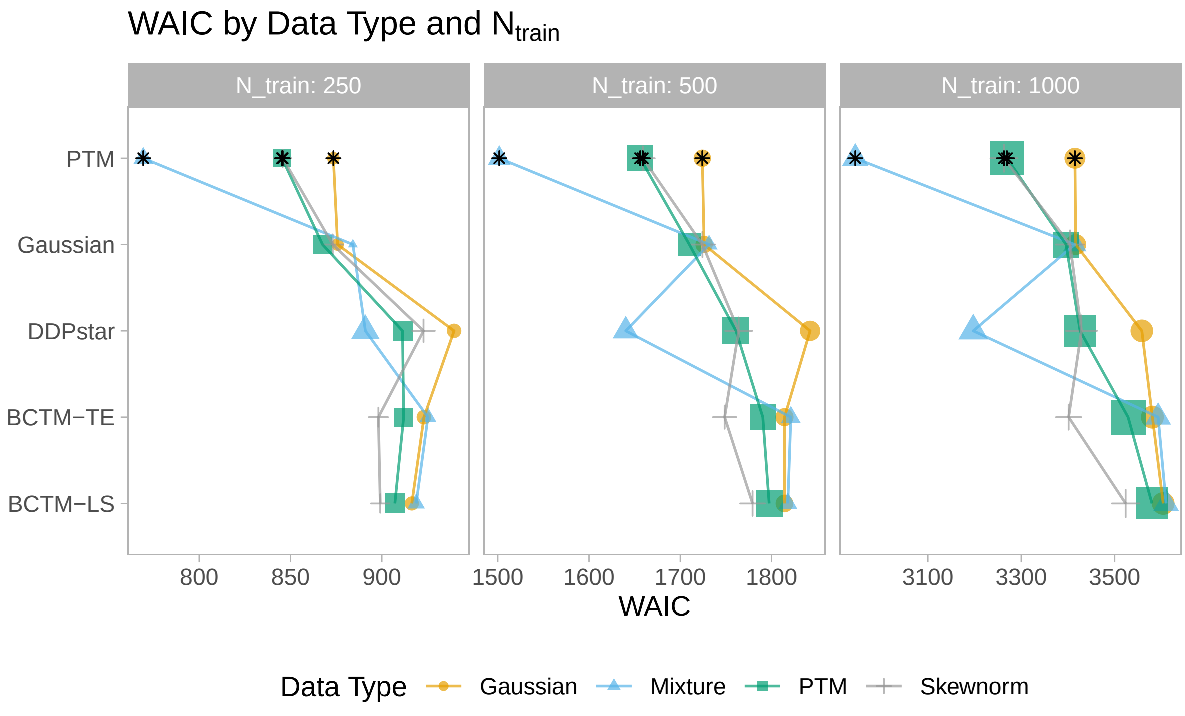


Figure I.2 Kullback-Leibler divergence. The plot shows the average of 100 replications with different seeds for pseudo-random number generation. The size of the point shapes is proportional to the standard deviation over these 100 replications. Within each panel and data type, a black star marks the best-performing observation. The top panel shows the plot without SBGP, since the inclusion of SBGP requires a wider scale.



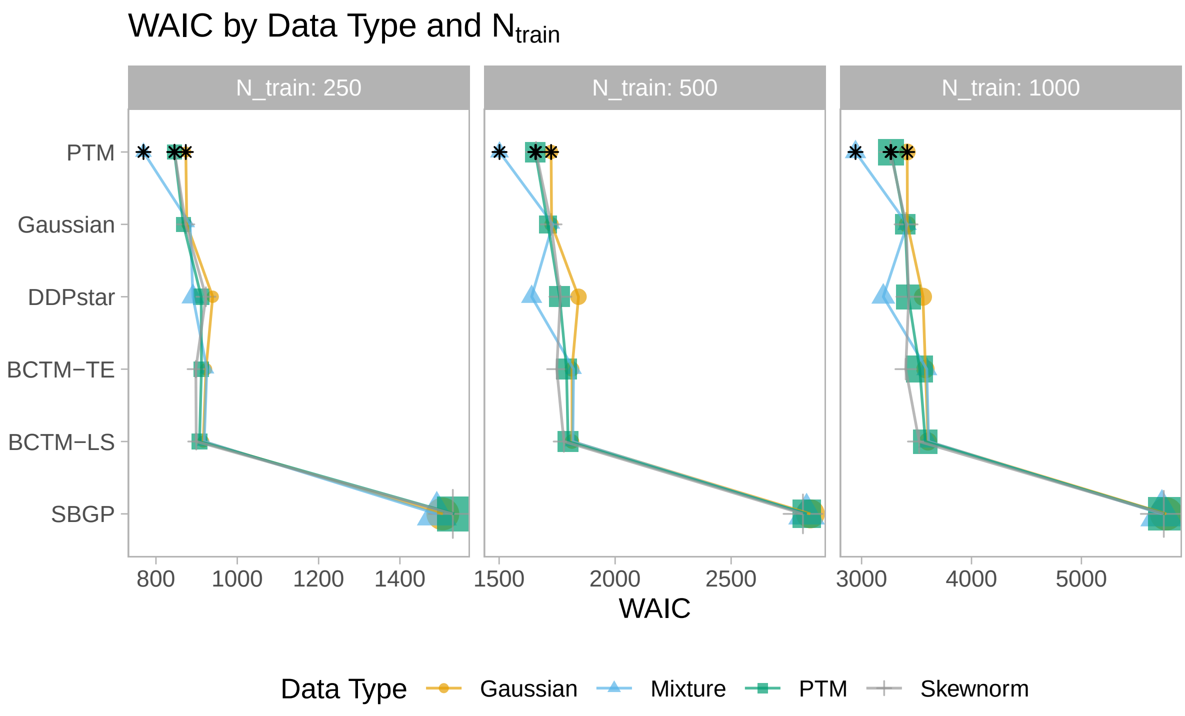
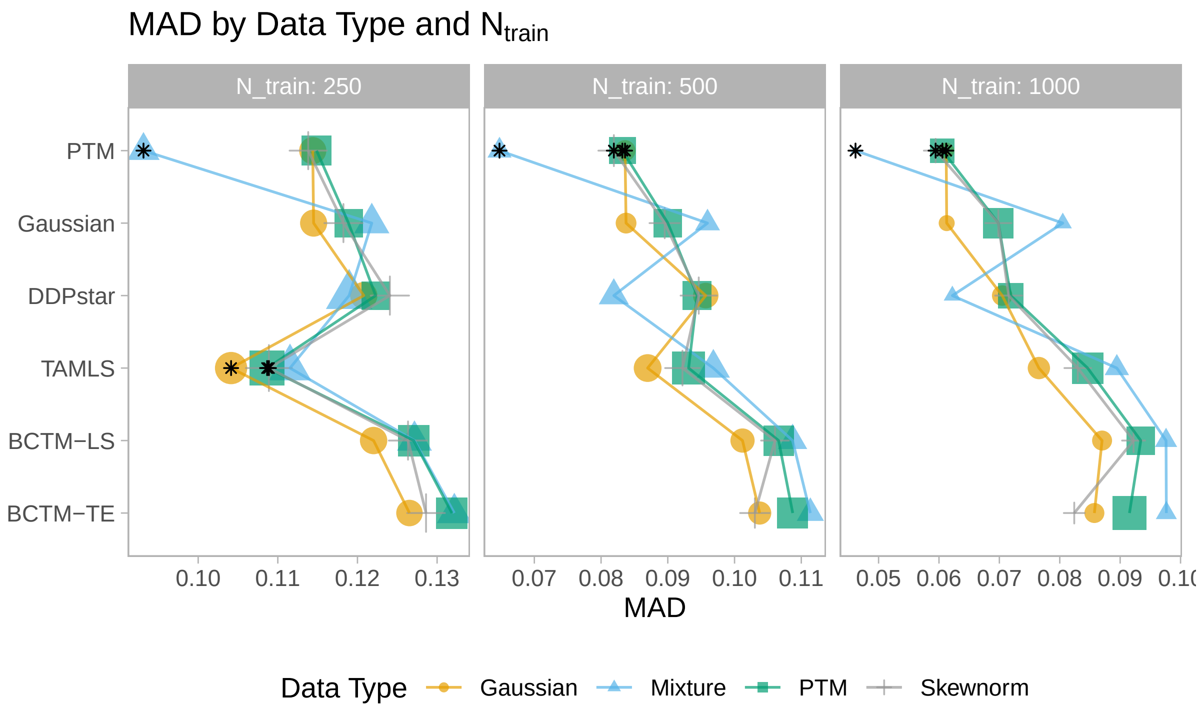


Figure I.3 WAIC. The plot shows the average of 100 replications with different seeds for pseudo-random number generation. The size of the point shapes is proportional to the standard deviation over these 100 replications. Within each panel and data type, a black star marks the best-performing observation. The top panel shows the plot without SBGP, since the inclusion of SBGP requires a wider scale.

Figure I.4 Continuous ranked probability score. The plot shows the average of 100 replications with different seeds for pseudo-random number generation. The size of the point shapes is proportional to the standard deviation over these 100 replications. Within each panel and data type, a black star marks the best-performing observation. The top panel shows the plot without SBGP, since the inclusion of SBGP requires a wider scale.



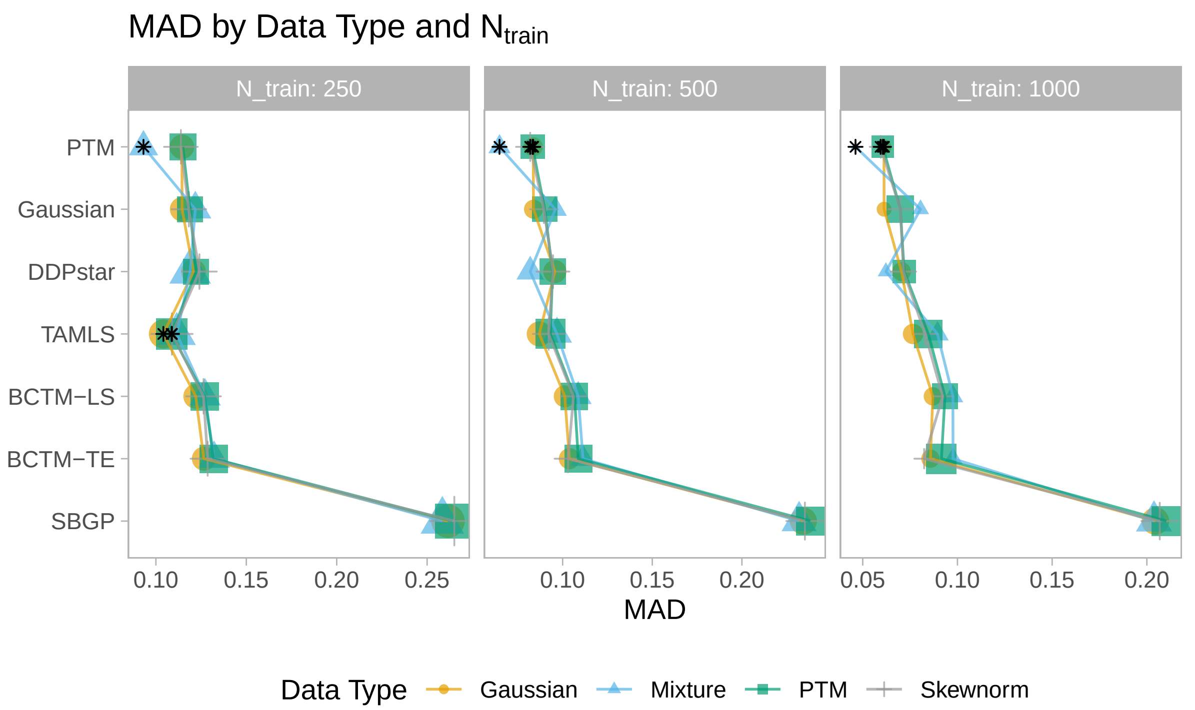


Figure I.5 Mean absolute difference in the CDF. The plot shows the average of 100 replications with different seeds for pseudo-random number generation. The size of the point shapes is proportional to the standard deviation over these 100 replications. Within each panel and data type, a black star marks the best-performing observation. The top panel shows the plot without SBGP, since the inclusion of SBGP requires a wider scale.

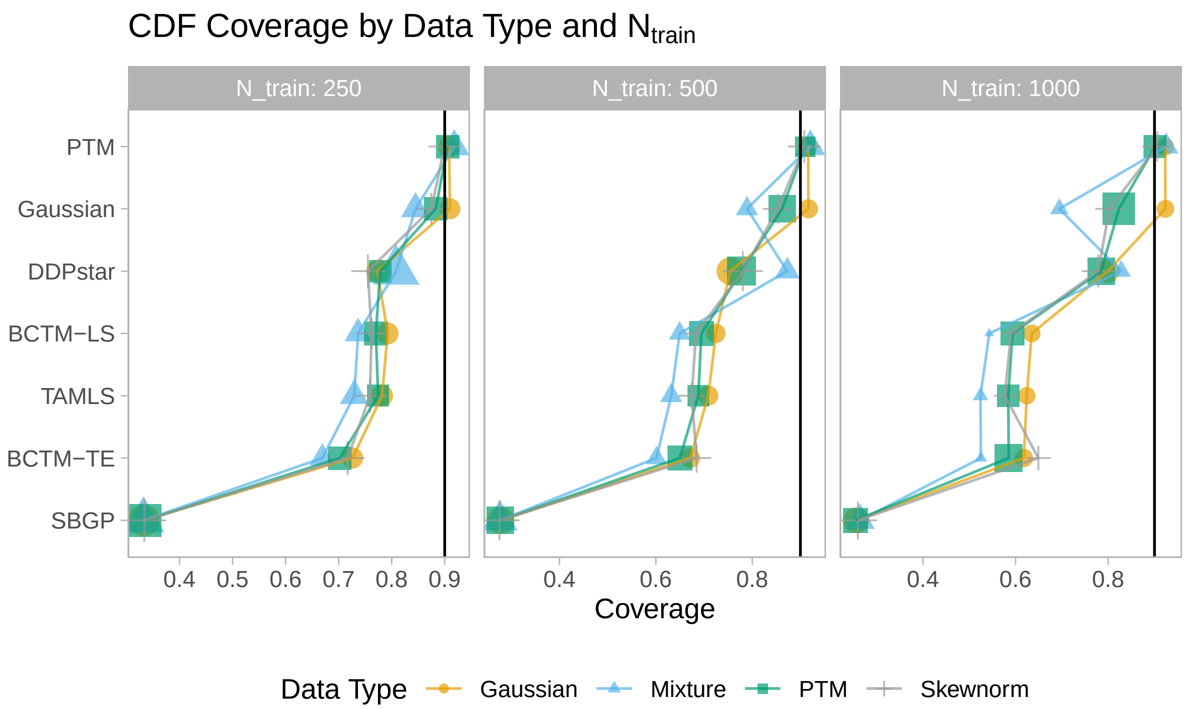


Figure I.6 Average pointwise coverage for 90% credible and confidence intervals of the CDF. Coverage within each replication is computed based on 5 000 random test observations from the target distribution. The plot shows the average of 100 replications with different seeds for pseudo-random number generation. The size of the point shapes is proportional to the standard deviation over these 100 replications. The vertical black line marks the nominal target level of .9.

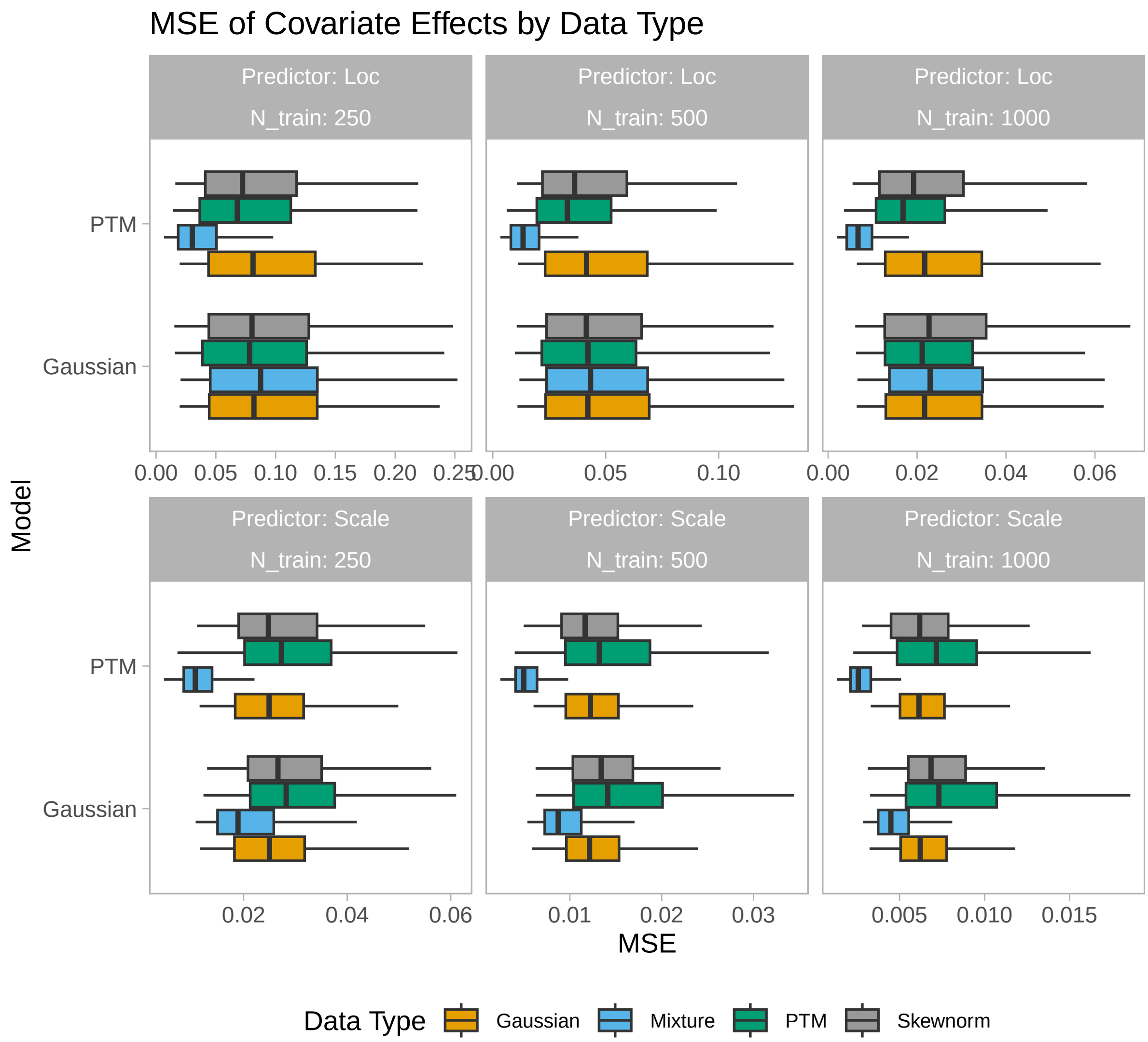


Figure I.7 Mean squared error for covariate effects. Each boxplot includes values from 100 replications and all four covariate functions, totaling 400 observations.

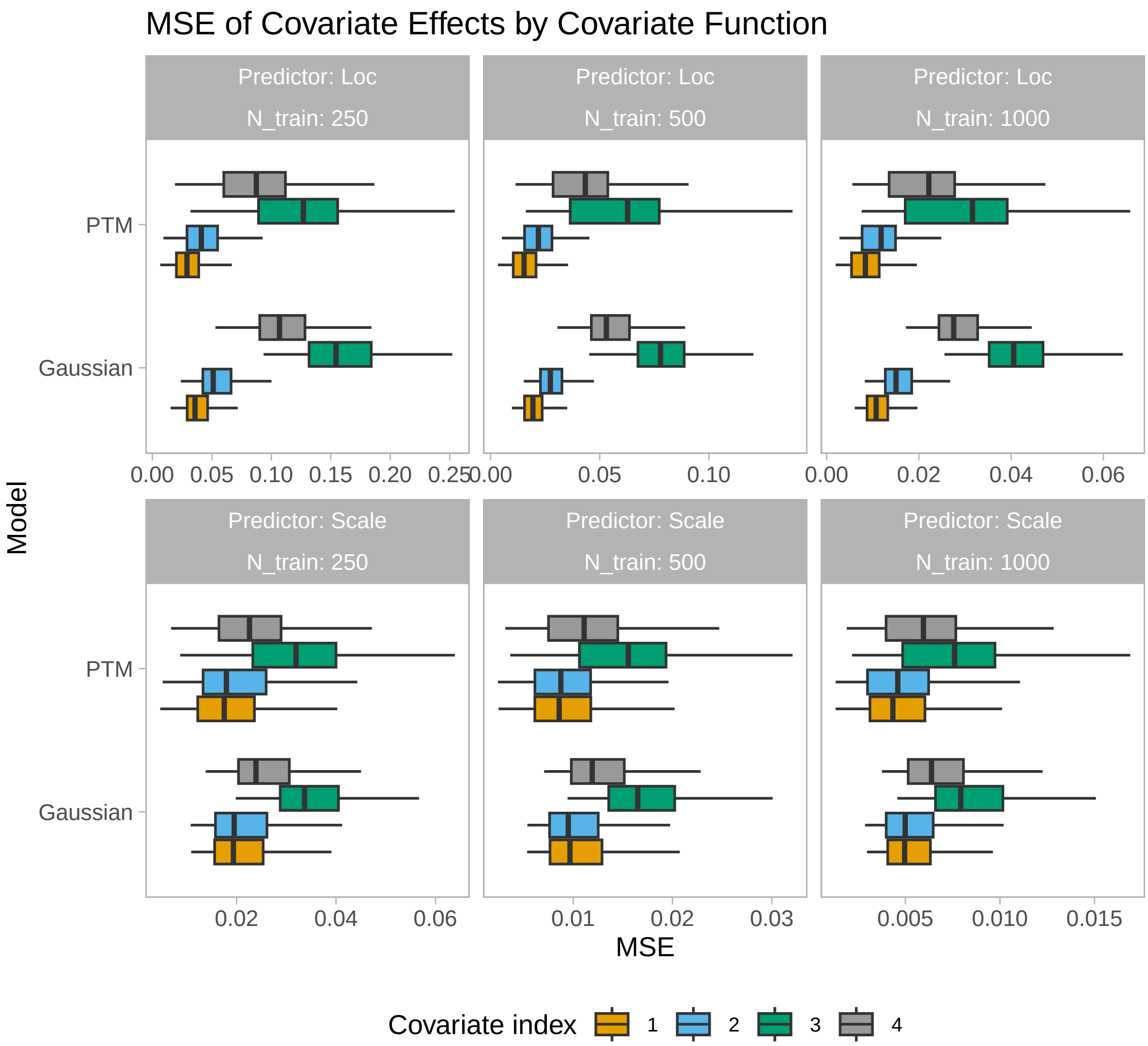


Figure I.8 Mean squared error for covariate effects. Each boxplot includes values from 100 replications and all four data generating mechanisms, totaling 400 observations.

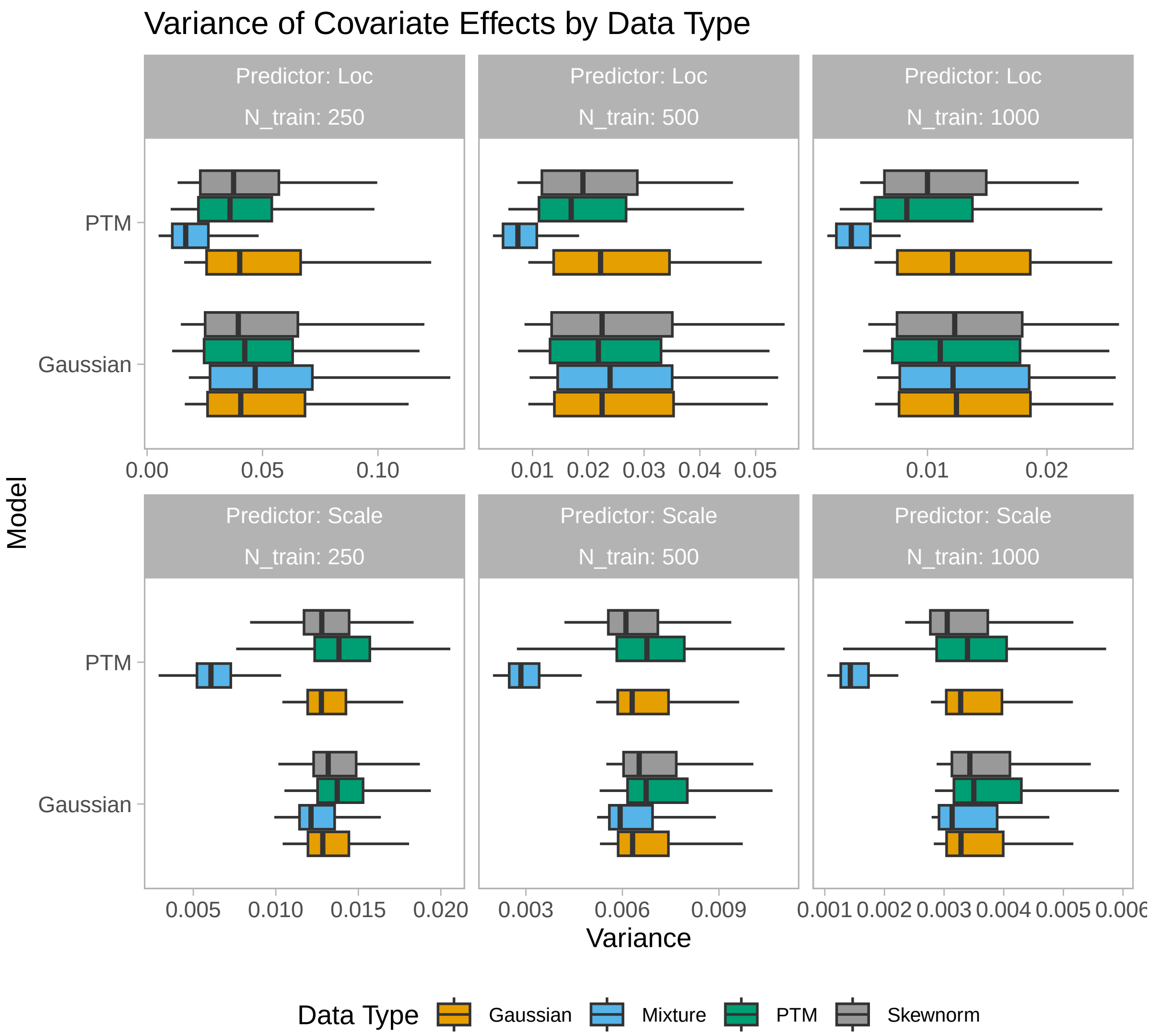


Figure I.9 Variance of covariate effects. Each boxplot includes values from 100 replications and all four covariate functions, totaling 400 observations.

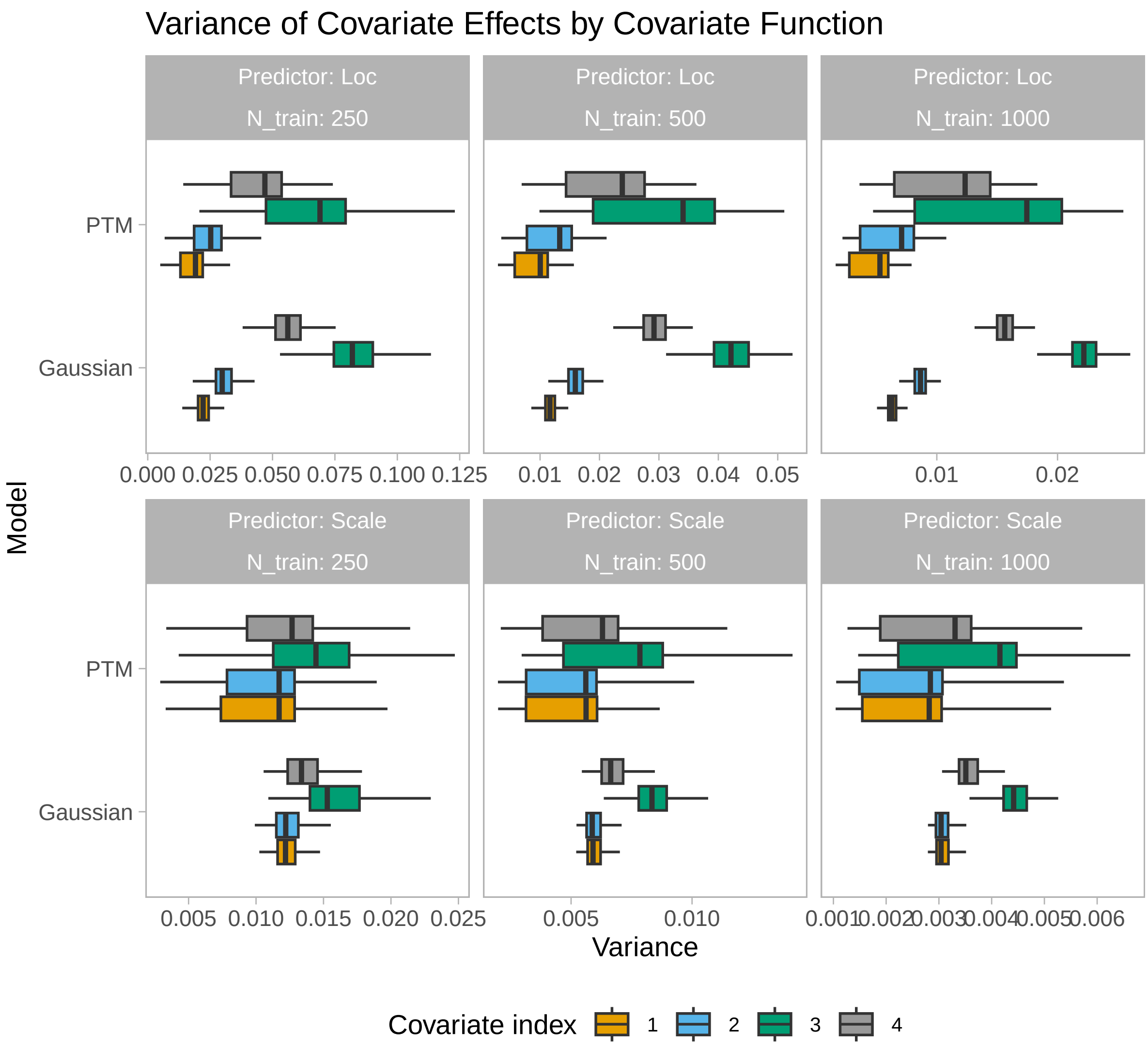


Figure I.10 Variance of covariate effects. Each boxplot includes values from 100 replications and all four data generating mechanisms, totaling 400 observations.

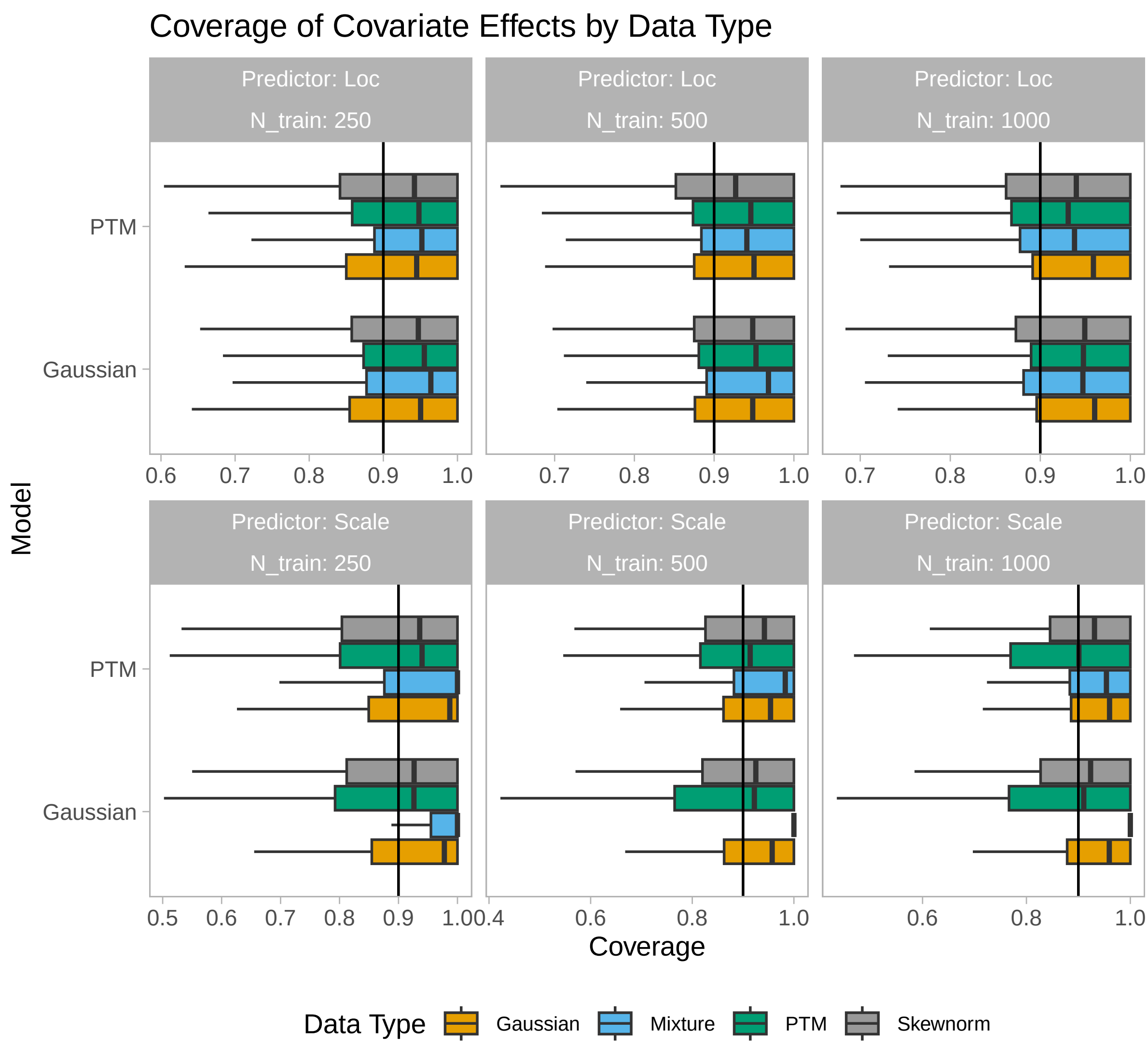


Figure I.11 Pointwise coverage for 90% credible and confidence intervals for covariate effects. Coverage within each replication is computed based on 5 000 random test observations from the target distribution. Each boxplot includes values from 100 replications and all four covariate functions, totaling 400 observations.

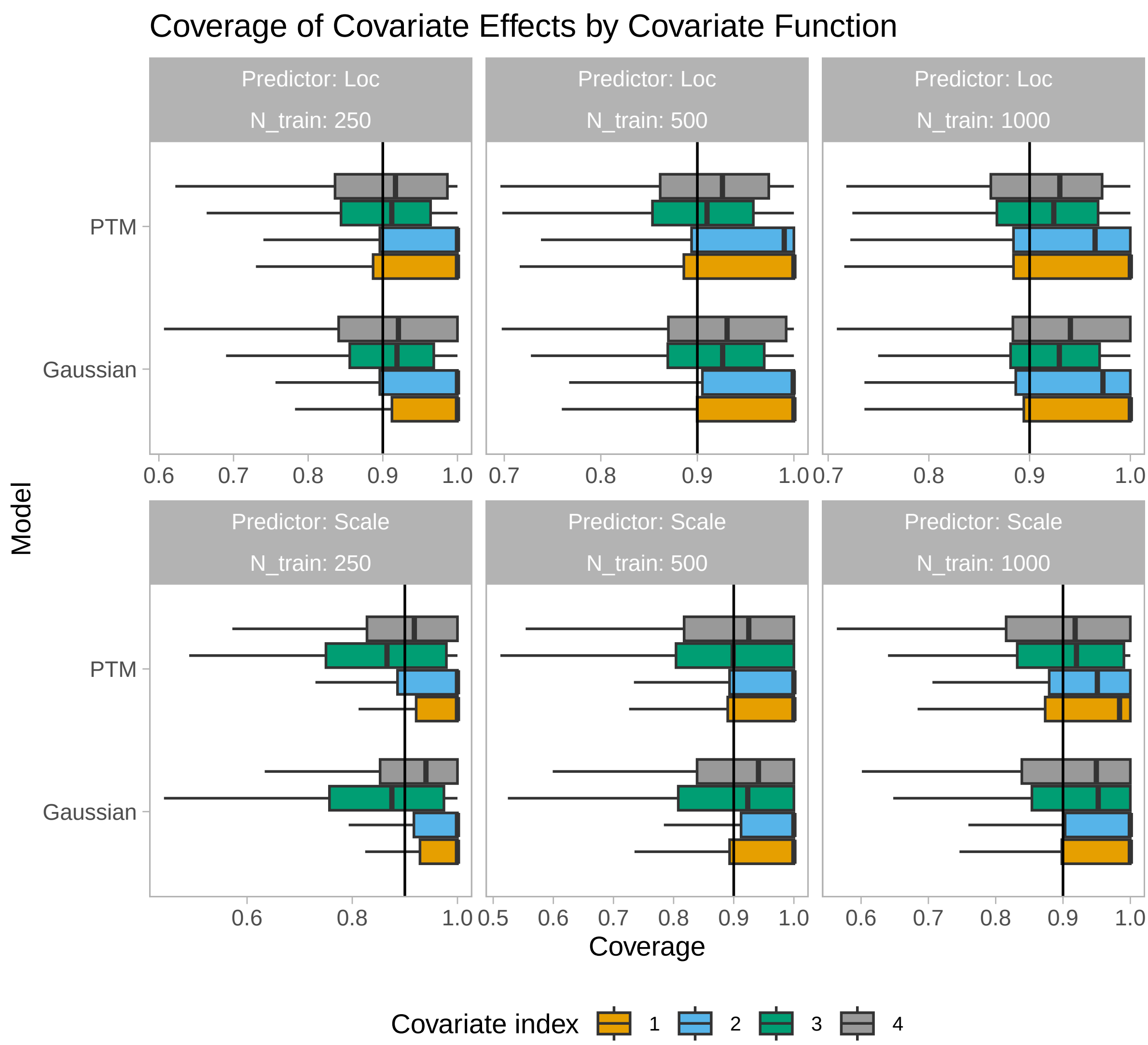


Figure I.12 Pointwise coverage for 90% credible and confidence intervals for covariate effects. Coverage within each replication is computed based on 5 000 random test observations from the target distribution. Each boxplot includes values from 100 replications and all four data generating mechanisms, totaling 400 observations.

I.6. Comparison of different inverse Gamma prior specifications

When considering the variance parameters τ^2 in the regularization priors used in our covariate effects, the inverse Gamma prior $\tau^2 \sim \mathcal{IG}(1, 0.001)$ used by us has a peak at small values and a heavy tail. When used as a prior for P-splines, this reflects skepticism about highly variable ("wiggly") functions, while still allowing the model to arrive at such functions if warranted by the data. When considering the precision $1/\tau^2$, an inverse Gamma prior with $a = 1$ and $b \to 0$ approaches a flat prior for the precision $1/\tau^2$. We ran the simulation study for the PTM in the two small-data scenarios ($N_{\text{obs}} \in \{250, 500\}$) using two other common settings for the inverse Gamma hyperpriors in the covariate effects:

- $a = b = 0.01$, which is a common prior that approaches a flat prior on $\ln \tau^2$ as $a, b \to 0$.
- $a = 1.0, b = 0.005$, which is another common variant of the "$a = 1$, b small" variety of inverse Gamma priors.

The two following tables show the results alongside the data underlying the results reported in the paper. Note that to keep the results comparable, we used only data from the $N_{\text{obs}} \in \{250, 500\}$ conditions for all models for these tables. While a small effect of the hyperprior is discernible, it does not lead to any change in conclusions.

Table I.4 Results for covariate effects. The results are averaged over all seeds, data types, sample sizes ($N_{\text{obs}} \in \{250, 500\}$), and over all four functions for the location and scale model parts, respectively. Coverages are reported for 90% credible intervals.

	Location Terms			Scale Terms		
Model	MSE ↓	Coverage	Width	MSE ↓	Coverage	Width
PTM (IG 0.01, 0.01)	0.065	0.914	0.553	0.027	0.929	0.374
PTM (IG 1, 0.005)	0.059	0.916	0.526	0.022	0.922	0.336
PTM (IG 1, 0.001)	0.054	0.916	0.496	0.018	0.905	0.291
Gaussian	0.070	0.922	0.579	0.020	0.912	0.318
DDPstar	0.082	0.907	0.606	0.035	0.451	0.173

Table I.5 Predictive performance and CDF coverage for the tested models. The results are averaged over all seeds, data types, and sample sizes ($N_{\text{obs}} \in \{250, 500\}$). For the KLD, MAD, and CDF, smaller values indicate a better predictive performance. Coverages are reported for 90% credible intervals.

					CDF CI	
Model	KLD ↓	CRPS ↓	WAIC ↓	MAD ↓	Coverage	Width
PTM (IG 0.01, 0.01)	0.113	0.826	1231	0.104	0.918	0.303
PTM (IG 1, 0.005)	0.101	0.821	1246	0.099	0.914	0.287
PTM (IG 1, 0.001)	0.094	0.826	1234	0.094	0.911	0.272
Gaussian	0.172	0.840	1299	0.104	0.867	0.290
DDPstar	0.227	0.859	1334	0.107	0.788	0.275
BCTM-LS	0.237	-	1356	0.116	0.727	0.268
BCTM-TE	0.246	-	1355	0.118	0.678	0.252
TAMLS	0.251	0.849	-	0.100	0.719	0.288
SBGP	-	1.492	-	0.253	0.898	0.898
QGAM	-	0.811	-	-	-	-

J. Application: Fourth Dutch Growth Study

This section includes additional information about the Fourth Dutch Growth Study application example.

J.1. Model specifications

- PTM: As described in the main text. The inverse smoothing parameters of the P-splines receive inverse gamma hyperpriors with concentration 1 and scale 0.001. We draw 10 000 samples each in four chains after a warmup of 5 000 iterations and use a thinning factor of 5, yielding a total of 8 000 samples.
- Gaussian: Gaussian location-scale model analogous to the PTM, changing the transformation function h to the identity function. We draw 20 000 samples each in four chains after a warmup of 5 000 iterations and use a thinning factor of 10, yielding a total of 8 000 samples.
- TAMLS: A transformation additive model (Siegfried et al., 2023)

$$\mathbb{P}(BMI \leq \text{bmi}_i) = \Phi(\sigma_i^{-1} h(\text{bmi}_i) - \mu_i) \tag{32}$$

$$\mu_i = \beta_0 + s(\text{age}_i) \tag{33}$$

$$\ln \sigma_i = g(\text{age}_i) \tag{34}$$

  with $\mu(\text{age}_i)$ and $\sigma(\text{age}_i)$ given by 20-parameter P-splines; $\sigma(\text{age}_i)$ does not include an intercept. The transformation function h is parameterized as an increasing Bernstein polynomial of order 10. Implementation was done by adapting the example code included in the R package `tram`, available on GitHub: https://github.com/cran/tram/blob/master/demo/stram.R
- SBGP: A semiparametric Bayesian Gaussian process model as proposed by Kowal and Wu (2024). The model is $g(\text{bmi}_i) = f_\theta(\text{age}_i) + \sigma\epsilon_i$ where $\epsilon_i \sim \mathcal{N}(0, 1)$ and $f_\theta \sim \mathcal{GP}(m_\theta, K_\theta)$. In the Gaussian process, m_θ is an unknown constant mean and K_θ is an isotropic Matérn covariance function with unknown variance, range, and smoothness parameters. We also tested a version of the model that includes a simple linear term $g(y_i) = \text{age}_i\beta + f_\theta(\text{age}_i) + \sigma\epsilon_i$. We found that the results improved when including this term and report the results for this model version. Computations are carried out using the code of the R package `SeBR` as provided by the authors in the supplementary materials of Kowal and Wu (2024). We save 5000 posterior predictive samples for each row of the test dataset.
- DDPstar: A density regression via Dirichlet process mixtures of normal structured additive regression models (Rodríguez-Álvarez et al., 2024). The means of the $l = 1, \ldots, L$ mixture components are modeled as 20-parameter P-splines $\mu(\text{age}_{i,l})$ using the default concentration 1 and scale 0.005. As Rodríguez-Álvarez et al. (2024) do in their example code, we use $L = 20$ mixture components, and otherwise stick to the defaults of their R library `DDPstar`,

which we use to carry out the computations. We draw 15 000 samples after a warmup of 5 000 iterations and use a thinning factor of 10, yielding a total of 1 500 samples.

- BCTM: Bayesian conditional transformation model (BCTM, Carlan et al., 2023) with standard normal reference distribution of the form

$$\mathbb{P}(BMI \leq \text{bmi}_i) = \Phi(h(\text{bmi}_i|\text{age}_i)),$$

where

$$h(\text{bmi}_i|\text{age}_i) = \left(\boldsymbol{a}(\text{bmi}_i)^\mathsf{T} \otimes \boldsymbol{b}(\text{age}_i)^\mathsf{T}\right)^\mathsf{T}\boldsymbol{\gamma}$$

is the BCTM's transformation function, with $\otimes$ denoting the Kronecker product. The bases $\boldsymbol{a}$ and $\boldsymbol{b}$ are chosen as 9-parameter B-spline bases, yielding a total of 81 parameters (including an intercept). The BCTM uses inverse gamma hyperpriors with concentration 1 and scale 0.001 for all inverse smoothing parameters of the tensor product terms. We apply MCMC via the No-U-Turn Sampler as available in the Python library `liesel-bctm` (Brachem et al., 2023). We draw 5 000 samples each in four chains after a warmup of 5 000 iterations, yielding a total of 20 000 samples. Since sampling is done exclusively with NUTS, we use no thinning.

- QGAM: A fully specified additive quantile regression model with P-splines for all four covariates, as available in the R package `qgam` (Fasiolo et al., 2021a; Fasiolo et al., 2021b). We fit QGAM for a grid of 25 evenly spaced probability levels in $\{0.005, \dots, 0.995\}$. Each quantile model is of the form

$$Q_{\text{bmi}}(\alpha|\text{age}_i) = s_\alpha(\text{age}_i)$$

where $s_\alpha(\text{age}_i)$ is a 20-parameter P-spline (including an intercept) and $Q_{\text{bmi}}(\alpha|\text{age}_i)$ is the conditional α-quantile of the response.

J.2. Additional results

Table J.1 shows predictive performance for all PTM variants that were run. Figure J.1 shows the corresponding quantile curves for the PTM variants. From both the plot and the table it becomes obvious that all models with $b = 7$ are practically equivalent, while the models with $b = 4$ struggle with accurately capturing the highest quantile of the data, and the model with $a = -4, b = 4$ and $\lambda = 0.1(b - a)$ yields notably more wiggly quantile curves than the other models.

Figure J.2 shows quantile curves for all models, including the PTM with $a = -4$, $b = 7$ and $\lambda = 0.1(b - a)$. We note a highly volatile quantile curve for the highest quantile in the BCTM and a visible underestimation of the lowest quantile in the Gaussian model.

Table J.2 contains diagnostic information about posterior MCMC samples for the Gaussian and PTM models. The quantities are computed using the Python library Arviz (Kumar et al., 2019), version 0.21.0.

- ESS (Bulk): Effective sample size for the bulk of the distribution, indicating how many independent draws the Markov chain is equivalent to in regions of high posterior mass.
- ESS (Tail): Effective sample size for the tail of the distribution, quantifying how well the sampler explores the extremes of the posterior.
- $\hat{R}$: The rank-normalized split $\hat{R}$ statistic, which compares within-chain to between-chain variance; values close to 1 indicate convergence, while values larger than 1.01 suggest problematic mixing (Gelman & Rubin, 1992).

The maximum $\hat{R}$ does not exceed 1.01 in any case, suggesting unproblematic mixing between the four sampled chains. The minimum effective sample sizes far exceed 400, suggesting sufficient information available in the posterior samples for all sampled quantities.

Table J.1 contains notes captured during MCMC sampling for the PTM and Gaussian models. The notes have the following meaning:

- `nan acceptance prob`: A numerical issue during sampling, leading to an invalid acceptance probability in a Metropolis-Hastings acceptance step. For Gaussian proposal distributions like the one used in iteratively weighted least squares proposals, large numbers here can indicate problems with the covariance matrix of the proposal distribution.
- `divergent transition`: Divergent transitions in NUTS sampling are numerical warnings that the Hamiltonian simulation could not accurately follow the posterior's geometry, often occurring in regions of high curvature or when parameters are poorly scaled. A few isolated divergences may not matter, but a persistent nonzero fraction (e.g. >1–2% of all transitions) indicates a serious problem with the sampler's validity (Betancourt, 2018).
- `maximum tree depth`: Maximum tree depth in NUTS sampling is a safeguard that limits how far the algorithm can explore a trajectory during each iteration. This note is emitted when the maximum tree depth is reached. It means that no U-turn was encountered in the simulated trajectory that generates the proposal, i.e. the leapfrog algorithm was not stopped early.

We observe minimal shares of divergent transitions and maximum tree depths, and virtually no NaN acceptance probabilities, showing no serious cause for concern.

Table J.1 Predictive performance for PTM model variants, based on 10-fold cross-validation. Smaller values for the Log Score and CRPS indicate better performance.

Model	a	b	λ	Log Score ↓	CRPS ↓
PTM	-4	7	$\lambda \to \infty$	625.69	0.365
PTM	-4	7	$\lambda = 0.1(b-a)$	625.69	0.365
PTM	-7	7	$\lambda \to \infty$	625.72	0.365
PTM	-7	7	$\lambda = 0.1(b-a)$	625.72	0.365
PTM	-4	4	$\lambda \to \infty$	638.65	0.364
PTM	-4	4	$\lambda = 0.1(b-a)$	640.25	0.364

Table J.2 Diagnostic information for the PTM and Gaussian models in the Fourth Dutch Growth Study application example. The values are averaged over all folds of the cross-validation. $\hat{R}$ values are based on four parallel chains.

Model	a	b	λ	Min. ESS (Bulk)	Min. ESS (Tail)	Max. $\hat{R}$
Gaussian	-	-	-	2 372	4 632	1.002
PTM	-4	4	$\to \infty$	1 510	3 407	1.003
PTM	-4	4	0.1(b-a)	2 685	5 362	1.002
PTM	-4	7	$\to \infty$	3 979	7 831	1.001
PTM	-4	7	0.1(b-a)	4 000	7 786	1.001

Table J.3 Notes captured during MCMC sampling for the PTM and Gaussian models in the Fourth Dutch Growth Study application example. The values are averaged over all folds of the cross-validation.

Model	a	b	λ	Note	Rel. freq.
Gaussian	-	-	-	nan acceptance prob	0
PTM	-4	4	$\to \infty$	divergent transition	0
PTM	-4	4	$\to \infty$	maximum tree depth	0
PTM	-4	4	0.1(b-a)	divergent transition	0
PTM	-4	4	0.1(b-a)	maximum tree depth	0
PTM	-4	7	$\to \infty$	divergent transition	0
PTM	-4	7	$\to \infty$	maximum tree depth	0
PTM	-4	7	0.1(b-a)	divergent transition	0
PTM	-4	7	0.1(b-a)	maximum tree depth	0

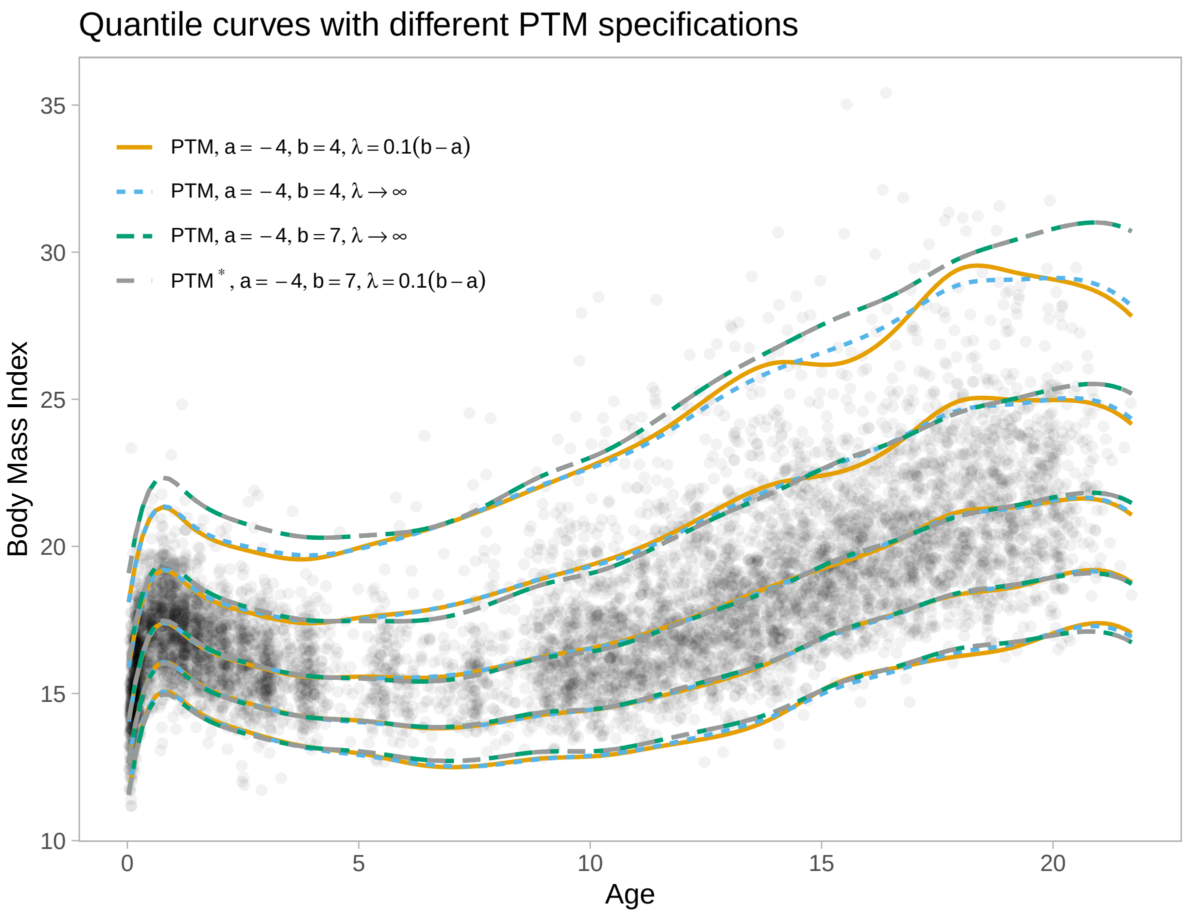


Figure J.1 Quantile curves at probability levels $0.01, 0.1, 0.5, 0.9, 0.99$ for PTM variants. The curves were produced with full-sample model runs.

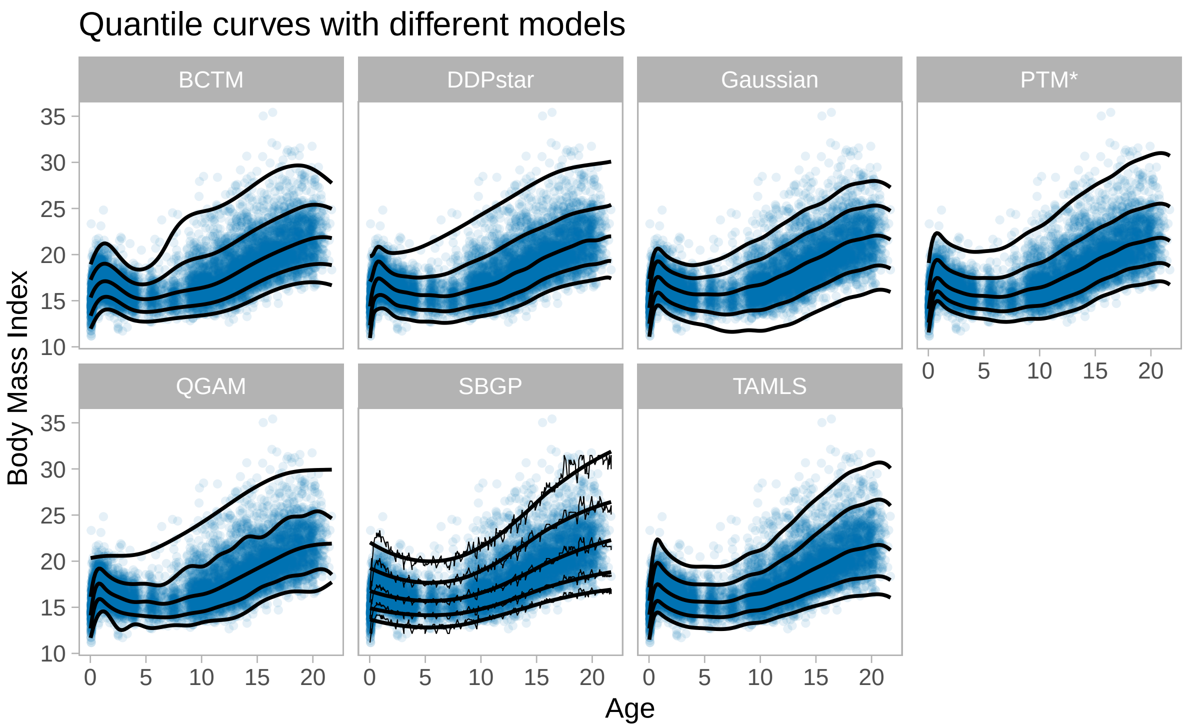


Figure J.2 Quantile curves at probability levels $0.01, 0.1, 0.5, 0.9, 0.99$ for different models. The curves were produced with full-sample model runs. For SBGP, the quantile curves were estimated based on a grid of 150 values for age in $[\min(age), \max(age)]$. For each grid point, we drew 5000 samples from the posterior predictive distribution. The fine, wiggly line shows the empirical quantiles of these samples. The overlaid smooth lines are cubic regression splines applied to the data (using the `ggplot2` default in `geom_smooth`).

K. Application: Framingham heart study

K.1. Model specifications

- PTM: As described in the main text. The inverse smoothing parameters of the P-splines receive inverse gamma hyperpriors with concentration 1 and scale 0.001. We draw 25 000 samples each in four chains after a warmup of 5 000 iterations and use a thinning factor of 5, yielding a total of 20 000 samples. For the random intercept, we use proposals from a No-U-Turn sampler instead of IWLS proposals, since we found that this improved sampling efficiency.
- Gaussian: Gaussian location-scale model analogous to the PTM, changing the transformation function h to the identity function. We draw 25 000 samples each in four chains after a warmup of 5 000 iterations and use a thinning factor of 5, yielding a total of 20 000 samples. For the random intercept, we use proposals from a No-U-Turn sampler instead of IWLS proposals, since we found that this improved sampling efficiency.
- TAMLS: A transformation additive model (Siegfried et al., 2023)

$$\mathbb{P}(Cholst \leq \text{cholst}_{ij}) = \Phi(\sigma_{ij}^{-1} h(\text{cholst}_{ij}) - \mu_{ij}) \tag{35}$$

$$\mu_{ij} = \beta_0 + s(\text{age}_i) + \text{sex}_i \beta_1 + \zeta_i \tag{36}$$

$$\ln \sigma_{ij} = g(\text{age}_i) + \text{sex}_i \gamma_1 \tag{37}$$

  with s and g given by 20-parameter P-splines; σ_{ij} does not include an intercept. On the location, we use an i.i.d. random intercept $\zeta_i \sim \mathcal{N}(0, \psi^2)$. The transformation function h is parameterized as an increasing Bernstein polynomial of order 10. We also include a model variant without random intercept. Implementation was done by adapting the example code included in the R package `tram`, available on GitHub: https://github.com/cran/tram/blob/master/demo/stram.R
- SBGP: A semiparametric Bayesian Gaussian process model as proposed by Kowal and Wu (2024). The model is $g(\text{cholst}_i) = \text{sex}_i \beta_1 + f_\theta(\text{age}_i)\beta_1 + \sigma\epsilon_i$ where $\epsilon_i \sim \mathcal{N}(0, 1)$ and $f_\theta \sim \mathcal{GP}(m_\theta, K_\theta)$. In the Gaussian process, m_θ is an unknown constant mean and K_θ is an isotropic Matérn covariance function with unknown variance, range, and smoothness parameters. We also tested a version of the model that includes a simple linear term $g(\text{cholst}_i) = \text{sex}_i \beta_1 + \text{age}_i \beta + f_\theta(\text{age}_i) + \sigma\epsilon_i$. We found that the results improved when including this term and report the results for this model version. Computations are carried out using the code of the R package `SeBR` as provided by the authors in the supplementary materials of Kowal and Wu (2024). We save 5 000 posterior predictive samples for each row of the test dataset.
- DDPstar: A density regression via Dirichlet process mixtures of normal structured additive regression models (Rodríguez-Álvarez et al., 2024). The means of the $l = 1, \ldots, L$ mixture components are modeled as $\mu_{i,l} = \beta_{0,l} + \text{sex}_i \beta_{1,l} + s(\text{age}_{i,l})$, where s are 20-parameter P-splines using the default concentration 1 and scale 0.005. As Rodríguez-Álvarez et al.

(2024) do in their example code, we use $L = 20$ mixture components, and otherwise stick to the defaults of their R library `DDPstar`, which we use to carry out the computations. We draw 15000 samples after a warmup of 5000 iterations and use a thinning factor of 10, yielding a total of 1500 samples. While `DDPstar` allows inclusion of random intercepts, it does not allow for prediction for unknown clusters in random intercept models, so we do not include a random intercept version.

- BCTM: Bayesian conditional transformation model (BCTM, Carlan et al., 2023) with standard normal reference distribution of the form

$$\mathbb{P}(Cholst \leq \text{cholst}_{ij}) = \Phi\big(h(\text{cholst}_{ij}|\text{age}_{ij}) + \text{sex}_i\gamma_2 + \zeta_i\big),$$

where

$$h(\text{cholst}_{ij}|\text{age}_{ij}) = \big(\boldsymbol{a}(\text{cholst}_{ij})^\top \otimes \boldsymbol{b}(\text{age}_{ij})^\top\big)^\top \boldsymbol{\gamma}_1$$

, with $\otimes$ denoting the Kronecker product. The bases $\boldsymbol{a}$ and $\boldsymbol{b}$ are chosen as 9-parameter B-spline bases, yielding a total of 81 parameters (including an intercept). The BCTM uses inverse gamma hyperpriors with concentration 1 and scale 0.001 for all inverse smoothing parameters of the tensor product terms. The term $\zeta_i \sim \mathcal{N}(0, \psi^2)$ is an i.i.d. random intercept. We also include a model variant without random intercept. We apply MCMC via the No-U-Turn Sampler as available in the Python library `liesel-bctm` (Brachem et al., 2023). We draw 5 000 samples each in four chains after a warmup of 5 000 iterations, yielding a total of 20 000 samples. Since sampling is done exclusively with NUTS, we use no thinning.

- QGAM: A fully specified additive quantile regression model with P-splines for all four covariates, as available in the R package `qgam` (Fasiolo et al., 2021a; Fasiolo et al., 2021b). We fit QGAM for a grid of 25 evenly spaced probability levels in $\{0.005, \dots, 0.995\}$. Each quantile model is of the form

$$Q_{\text{cholst}}(\alpha|\text{age}_{ij}) = s_\alpha(\text{age}_{ij}) + \text{sex}_i\beta_1$$

where $s_\alpha(\text{age}_i)$ is a 20-parameter P-spline (including an intercept) and $Q_{\text{cholst}}(\alpha|\text{age}_{ij})$ is the conditional α-quantile of the response. While `qgam` allows inclusion of random intercepts, we found model fitting to be computationally infeasible, so we do not include a random intercept version.

K.2. Additional results

Figure K.1 shows quantile curves for all models. Table K.1 contains diagnostic information about posterior MCMC samples for the Gaussian and PTM models. The quantities are computed using the Python library Arviz (Kumar et al., 2019), version 0.21.0. The maximum $\hat{R}$ exceeds 1.01 for the PTMs with random intercept, suggesting a problem with convergence. When we inspected these models more closely, we found that the posterior distribution of the random intercept term closely resembles the posterior estimate for $f_R(r)$ in the corresponding model without random intercept. This could indicate that the transformation function estimate and the random intercept capture similar information, leading to challenging identification and hence problematic convergence.

The minimum effective sample sizes exceed 400 for all models, suggesting sufficient information available in the posterior samples for all sampled quantities.

Table K.2 contains notes captured during MCMC sampling for the PTM and Gaussian models. We observe minimal shares of divergent transitions and maximum tree depths, and virtually no NaN acceptance probabilities, showing no serious cause for concern.

Table K.1 Diagnostic information for the PTM and Gaussian models in the Framingham Heart Study application example. The values are averaged over all folds of the cross-validation. $\hat{R}$ values are based on four parallel chains.

Model	Minimum ESS (Bulk)	Minimum ESS (Tail)	Maximum $\hat{R}$
PTM (lin)	4 569	8 698	1.001
PTM (lin) +RI	1 501	1 006	1.004
PTM (nonlin)	2 247	4 380	1.002
PTM (nonlin) +RI	1 456	2 005	1.002
Gaussian (lin)	13 977	16 360	1.000
Gaussian (lin) +RI	2 207	4 327	1.002
Gaussian (nonlin)	4 975	8 633	1.001
Gaussian (nonlin) +RI	2 264	4 316	1.002

Table K.2 Notes captured during MCMC sampling for the PTM and Gaussian models in the Framingham Heart Study application example. The values are averaged over all folds of the cross-validation.

Model	Note	Rel. freq.
PTM (lin)	divergent transition	0
PTM (lin)	maximum tree depth	0
PTM (lin) +RI	divergent transition	0
PTM (lin) +RI	maximum tree depth	0
PTM (nonlin)	divergent transition	0
PTM (nonlin)	maximum tree depth	0
PTM (nonlin) +RI	divergent transition	0
PTM (nonlin) +RI	maximum tree depth	0
PTM (nonlin) +RI	nan acceptance prob	0
Gaussian (lin)	nan acceptance prob	0
Gaussian (lin) +RI	divergent transition	0
Gaussian (lin) +RI	nan acceptance prob	0
Gaussian (nonlin)	nan acceptance prob	0
Gaussian (nonlin) +RI	divergent transition	0
Gaussian (nonlin) +RI	nan acceptance prob	0

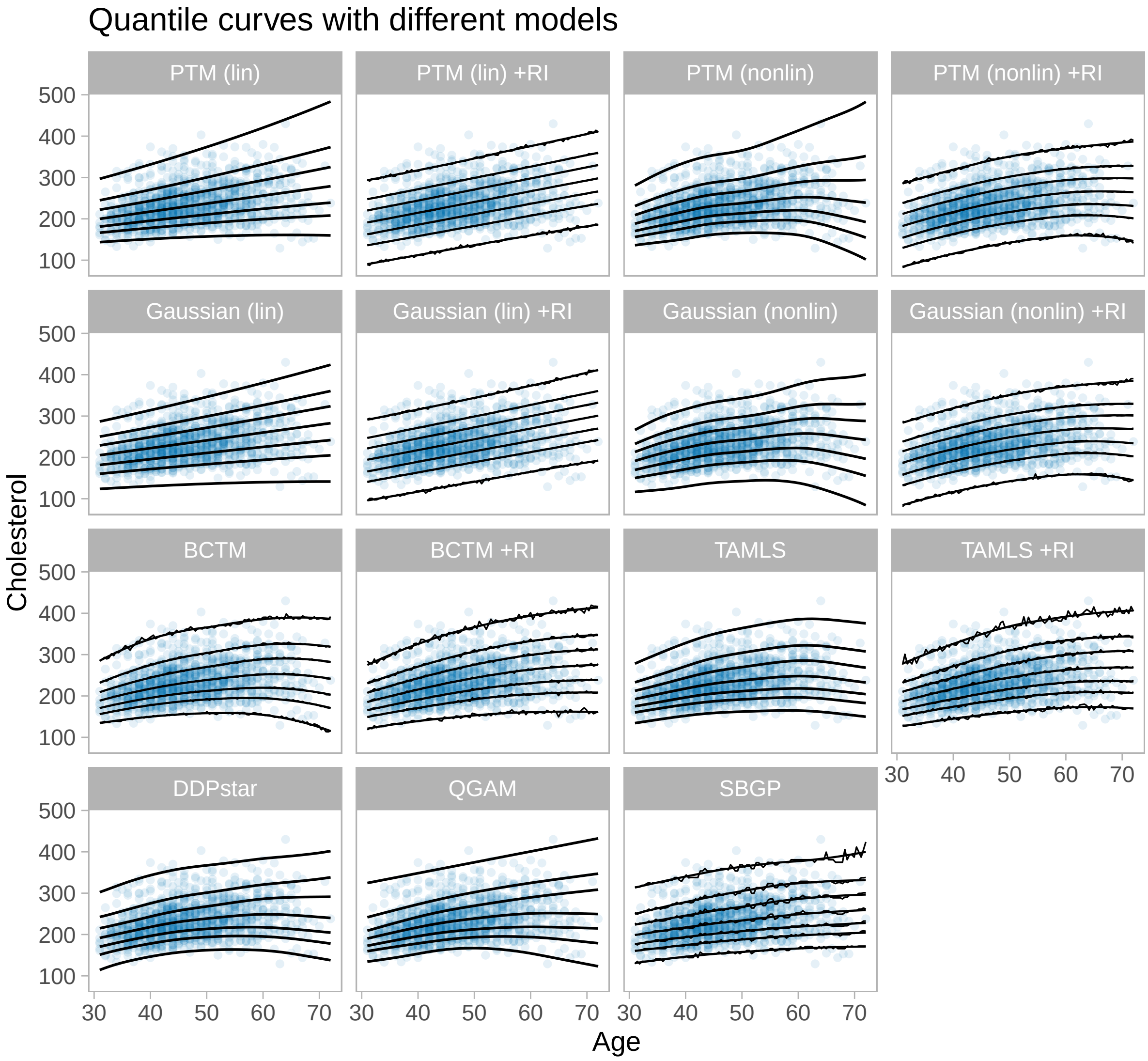


Figure K.1 Quantile curves at probability levels $0.01, 0.1, 0.5, 0.9, 0.99$ for different models. The curves were produced with full-sample model runs. The models labeled with +RI include an i.i.d. random intercept term for each patient. For BCTM, SBGP and the random intercept models, the quantile curves were estimated based on a grid of 150 values for age in $[\min(age), \max(age)]$. For each grid point, we drew samples from the posterior predictive distribution, drawing a random intercept term from $\mathcal{N}(0, \psi^2)$. For the random intercept variance ψ^2, we use the variance point estimate (TAMLS) or draws from the variance parameter's posterior distribution (Bayesian models). The fine, wiggly line shows the empirical quantiles of these samples. The overlaid smooth lines are cubic regression splines applied to the data (using the `ggplot2` default in `geom_smooth`).

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